

**An-Najah National University**  
**Computer Engineering Department**

**Course: Database Systems**

**Final Project Report**  
**MixMart Online Store**  
**Online Shopping Store Database Project**

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# 1. Introduction & Problem Statement

Online shopping systems depend on organized data, accurate stock tracking, and reliable order processing. Without a well-designed database, a store may face repeated customer information, incorrect product quantities, missing order details, and difficulty following customer requests.

Mix Mart Online is a Java Swing desktop application and MySQL database system designed to manage the main operations of an online shopping store. The system connects customers, products, carts, orders, payments, shipments, reviews, selling requests, and chat messages in one organized structure.

The project solves the problem of managing store operations manually or through unorganized files. Instead of storing all information in one large table, the system separates data into related tables and connects them using primary keys and foreign keys. This improves data integrity, reduces redundancy, and makes the application easier to update and maintain.

## 2. Executive Summary

MixMart Online Store is a Java Swing application connected to a MySQL database. It was developed for the Database Systems course to show how database design can support a real shopping workflow. The project includes customer features and admin features, so the database is used directly in daily operations such as browsing products, creating orders, managing stock, handling reviews, approving selling requests, and sending chat messages.

## 3. System Architecture & Technologies

Mix Mart Online follows a simple layered structure. The Java Swing interface handles windows and user actions, while DAO classes handle SQL operations. This separation makes the project easier to read, test, and maintain.

- Frontend: Java Swing desktop interface for forms, buttons, dialogs, product cards, cart sidebar, and admin panels.
- Backend Database: MySQL database named `online_shopping_store_database`.
- Connectivity: JDBC is used to connect the Java application with MySQL and execute SQL statements.
- Data Access Layer: DAO classes separate database logic from GUI code, such as CustomerDAO, ProductDAO, OrderDAO, ChatDAO, and AdminStatsDAO.
- Assets: Product images are loaded from file paths stored in the database and displayed inside the product cards.

| Tool / Technology     | Use in Project                        | Reason for Use                                                             |
|-----------------------|---------------------------------------|----------------------------------------------------------------------------|
| Java                  | Main application programming language | Used to build the desktop application logic.                               |
| Java Swing            | Graphical user interface              | Used for forms, buttons, panels, product cards, cart sidebar, and dialogs. |
| MySQL                 | Relational database management system | Used to create and manage the project database.                            |
| JDBC                  | Database connectivity                 | Used to connect Java with MySQL and execute SQL statements.                |
| DAO Classes           | Application structure                 | Used to separate database operations from GUI code.                        |
| SQL                   | Database operations                   | Used for DDL and DML statements.                                           |
| Product Images Folder | Image assets                          | Used to display product images inside product cards.                       |

## 4. Business Rules & Role-Based Access Control

The system applies clear rules based on the user type. Mix Mart Online has two main roles: Customer and Admin. Each role has different permissions to keep the workflow organized and protect customer-only or admin-only operations.

- Customers can register, log in, browse products, use cart and wishlist, checkout, view orders, rate products, submit selling requests, and chat with the admin.
- The admin can manage products, view all orders, check statistics, view reviews, process selling requests, and reply to customer messages.
- Products with zero stock cannot be added to the cart, and cart quantity cannot exceed available stock.
- Product reviews are allowed only after the related order is shipped or delivered.
- Pending selling requests can be edited or deleted by the customer, while the admin can approve or reject them.
- The admin can reply to a specific customer or send one broadcast message to all customers.

| User Type | Main Role                                                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Customer  | Registers, logs in, browses products, uses cart/wishlist, checks out, reviews products, submits selling requests, and chats with admin. |
| Admin     | Manages products, orders, reviews, statistics, selling requests, and customer chat messages.                                            |

## 5. Database Design Overview

The database contains related tables instead of one large table. Product categories are stored in Category and referenced by Product. Orders are stored in Orders, while the products inside each order are stored in OrderItem. This structure reduces repeated data and makes the database easier to maintain.

## 6. Functional Requirements

The system provides the main functions needed for a simple online shopping store. Customers can register, log in, browse products, search and filter items, add products to the cart or wishlist, complete checkout, view and confirm orders, rate products and the website, submit selling requests, and chat with the admin.

The admin side includes product management, order viewing, statistics, review viewing, selling request approval or rejection, and customer chat support. These requirements were implemented to make the database part of real user operations instead of being only static tables.

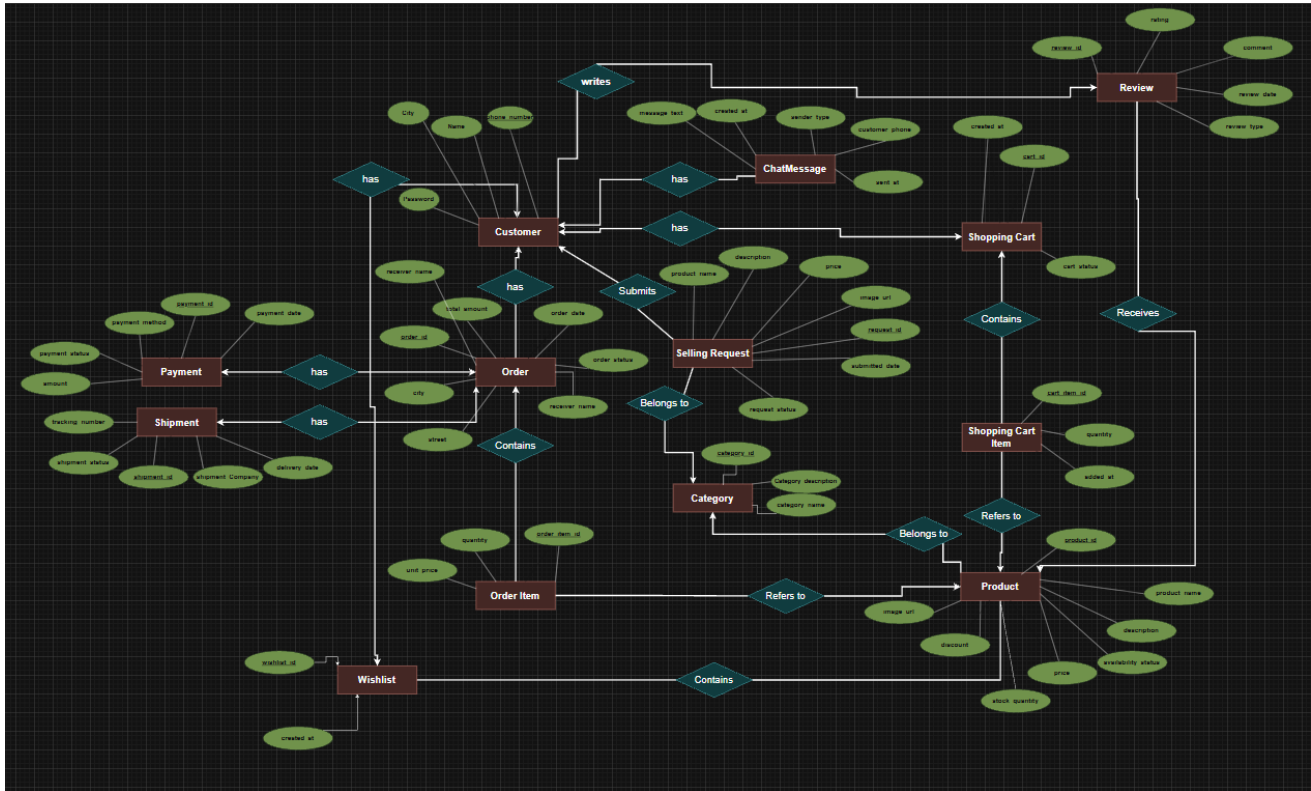
## 7. Non-Functional Requirements

The system was designed to be simple, organized, and easy to use. The Java Swing interface allows users to perform the main store operations without writing SQL queries manually. DAO classes were used to keep database operations separated from the GUI code, which makes the project easier to maintain.

The database supports data integrity using primary keys, foreign keys, unique constraints, and check constraints. The application also validates important input such as names, phone numbers, prices, quantities, stock availability, and card payment information. Basic login checking is included, while password hashing can be added in the future for stronger security.

## 8. ER Diagram Explanation

The ER diagram shows the main entities and relationships in the store database. Customer and Product are central entities. A customer can have a cart, create orders, write reviews, submit selling requests, use a wishlist, and send chat messages. Products belong to categories and can appear in carts, orders, reviews, and wishlists.



In the diagram, Customer is one of the central entities. A customer can have a shopping cart, create orders, write reviews, add wishlist items, submit selling requests, and send chat messages. Product is also central because it belongs to a category, appears in cart items and order items, can be reviewed, and can appear in wishlists. Order connects customers with purchased products and is linked with payment and shipment information.

## 9. Relational Schema

After designing the ERD, the entities were converted into relational tables. The following schema summarizes the main tables, their primary keys, and important foreign keys.

| Table            | Primary Key   | Main Foreign Keys                                 | Purpose                                   |
|------------------|---------------|---------------------------------------------------|-------------------------------------------|
| Customer         | phone_number  | -                                                 | Stores customer account data.             |
| Category         | category_id   | -                                                 | Stores product categories.                |
| Product          | product_id    | category_id -> Category                           | Stores product details.                   |
| ShoppingCart     | cart_id       | customer_phone -> Customer                        | Stores one cart for each customer.        |
| ShoppingCartItem | cart_item_id  | cart_id -> ShoppingCart,<br>product_id -> Product | Stores products inside a cart.            |
| Orders           | order_id      | customer_phone -> Customer                        | Stores order header and delivery details. |
| OrderItem        | order_item_id | order_id -> Orders, product_id -> Product         | Stores products inside an order.          |

|                |             |                                                                             |                                                  |
|----------------|-------------|-----------------------------------------------------------------------------|--------------------------------------------------|
| Payment        | payment_id  | order_id -> Orders                                                          | Stores payment method and status.                |
| Shipment       | shipment_id | order_id -> Orders                                                          | Stores shipping company and tracking data.       |
| Review         | review_id   | customer_phone -> Customer,<br>product_id -> Product, order_id<br>-> Orders | Stores product and website reviews.              |
| Wishlist       | wishlist_id | customer_phone -> Customer,<br>product_id -> Product                        | Stores wishlist products.                        |
| SellingRequest | request_id  | customer_phone -> Customer,<br>category_id -> Category                      | Stores used-product requests.                    |
| ChatMessage    | message_id  | customer_phone -> Customer                                                  | Stores chat messages between customer and admin. |

## 10. Table-by-Table Database Description

### Customer

The Customer table stores the main account information for each user. The phone number is used as the primary key because the system uses it for login and customer identification. This table includes important details such as the customer name, city, password, and account creation date.

### Category

The Category table organizes products into clear groups such as Computer Parts, Used Products, Mobile Phones, Home Electronics, Books, Clothes, and other store sections. Each product is linked to one category using category\_id as a foreign key.

### Product

The Product table stores all items available in the store. It includes the product name, description, price, image path, discount status, availability status, category, stock quantity, and creation date. This table is important because many other parts of the system depend on product information.

### ShoppingCart

The ShoppingCart table stores the active cart for each customer. Each cart is connected to one customer using customer\_phone. This makes it possible for every logged-in customer to have their own shopping cart.

### ShoppingCartItem

The ShoppingCartItem table stores the products added to a customer's cart. It connects ShoppingCart with Product and stores the selected quantity and the date the item was added. This table handles the many-to-many relationship between carts and products.

## Orders

The Orders table stores the main information about each order. It includes the customer phone, total amount, order date, order status, receiver name, receiver phone, city, and street. This table represents the order as a whole before looking at the individual products inside it.

## OrderItem

The OrderItem table stores the products included in each order. It connects Orders with Product and stores the quantity and unit price. Keeping the unit price here is important because it preserves the product price at the time of purchase, even if the product price changes later.

## Payment

The Payment table stores payment details for each order, such as payment method, payment status, payment date, and amount. Each order has one payment record, which keeps payment information separate from the general order details.

## Shipment

The Shipment table stores delivery information for each order. It includes the shipment company, shipment status, tracking number, and delivery date. Separating shipment data makes it easier to update order delivery information without changing the main order record.

## Review

The Review table stores both product reviews and website reviews. It includes the customer phone, product ID if the review is related to a product, order ID if needed, rating, comment, review date, and review type. The review\_type field helps the system know whether the review is for a product or for the website.

## Wishlist

The Wishlist table stores products that customers want to save for later. It connects Customer with Product using customer\_phone and product\_id. This table helps customers keep a list of products they are interested in without adding them to the cart.

## SellingRequest

The SellingRequest table stores used products submitted by customers. It includes the customer phone, product name, description, price, image path, request status, and submission date. The admin can approve or reject these requests. Approved requests can later appear as used products in the store.

## ChatMessage

The ChatMessage table stores messages between customers and the admin. Each message belongs to a customer and includes sender\_type to show whether the message was sent by the customer or by the admin. This table also stores the message text and the sending time, which allows the system to display the chat history in order.

## 11. Normalization Discussion

The database was organized mainly up to 3NF. In 1NF, repeated product lists were removed by using OrderItem and ShoppingCartItem. In 2NF, product information was kept in Product instead of being repeated in relationship tables. In 3NF, category details, payment details, shipment details, and review details were separated into their own tables. This reduces redundancy and avoids common update, insertion, and deletion problems

## 12. SQL Script and Data Operations

The SQL script includes DDL statements for creating the database schema and DML statements for inserting sample data and supporting project operations. The application uses SELECT, INSERT, UPDATE, and DELETE through DAO classes. The database also uses constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, DEFAULT values, and TIMESTAMP columns.

## 13. Java Application Architecture

The Java application was organized around the GUI class and several DAO classes. The GUI class builds the interface and handles button events. DAO classes contain the SQL operations. This design makes the project easier to manage because SQL statements are separated from the interface code.

| Class             | Responsibility                                                                                |
|-------------------|-----------------------------------------------------------------------------------------------|
| DBConnection      | Creates and returns the database connection using JDBC.                                       |
| CustomerDAO       | Handles customer registration, login, and fetching customer names.                            |
| ProductDAO        | Loads products, adds products, updates products, hides/deletes products, and checks stock.    |
| CartDAO           | Handles cart item insertion, removal, and quantity updates.                                   |
| OrderDAO          | Creates orders, loads customer orders, confirms payment and shipment, and loads admin orders. |
| ReviewDAO         | Handles product reviews, website reviews, and admin review viewing.                           |
| WishlistDAO       | Adds, loads, and removes wishlist products.                                                   |
| SellingRequestDAO | Submits, edits, deletes, accepts, and rejects selling requests.                               |
| ChatDAO           | Sends messages, loads messages, loads customers, and sends broadcast messages.                |
| AdminStatsDAO     | Calculates statistics for admin dashboard.                                                    |

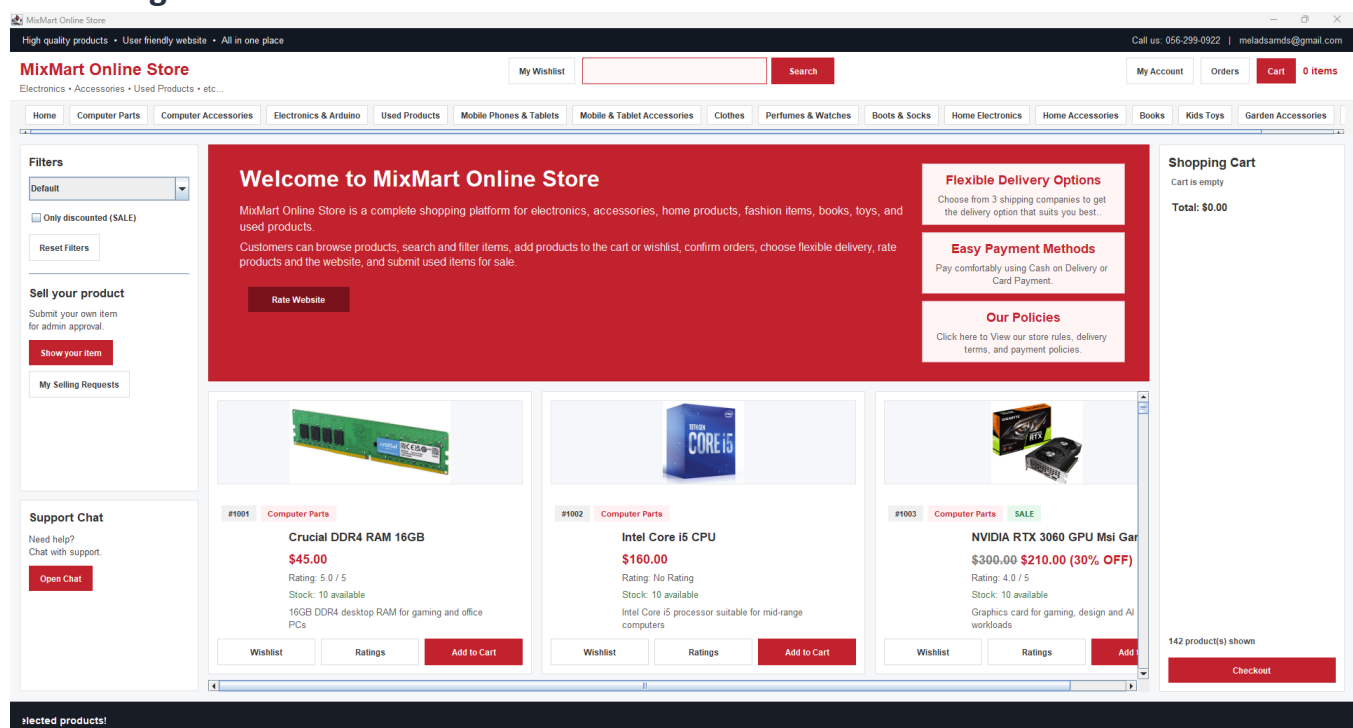


## Note: Error and warning messages are shown later in the 18. Testing and Validation section.

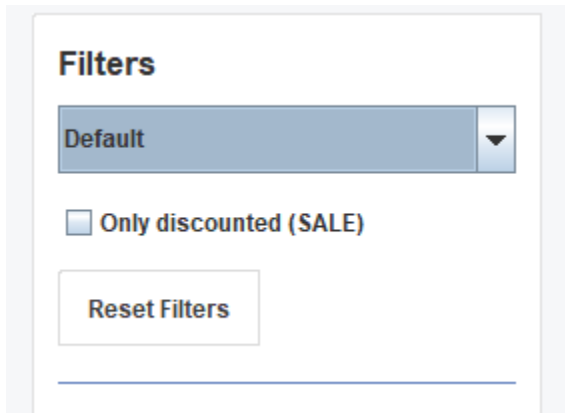
### 14. GUI Design and User Interface Screens

This section contains placeholder areas for screenshots from the final Java Swing application. The screenshots should be replaced with real images before final submission. Each screenshot should show a clear part of the system so the report can support the live demo and presentation.

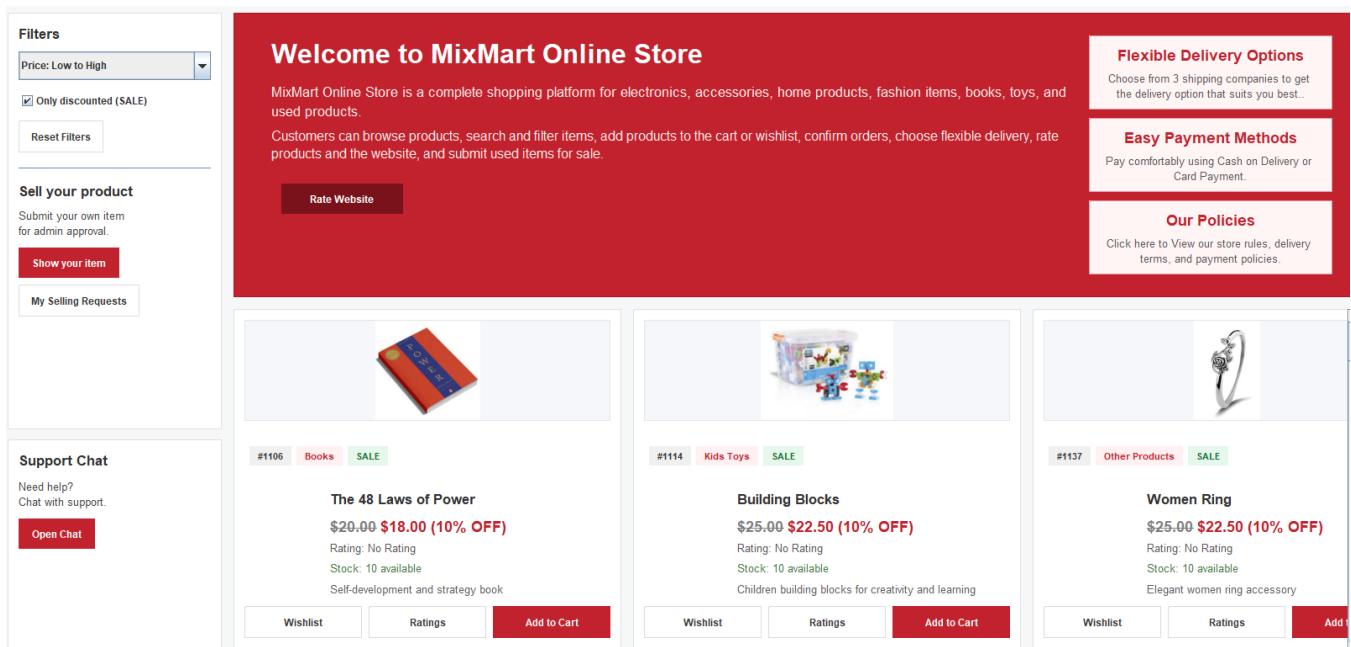
#### Home Page and Products



User here are considered as Guest and can search for the items they wants and can watch items but cant use any features like open chat or show your item or even add to cart or Wishlist but they can see the ratings of the item cant rate website or prouduct until they register or login, they can use filters



We can apply 2 filters discount and any other and they will be applied

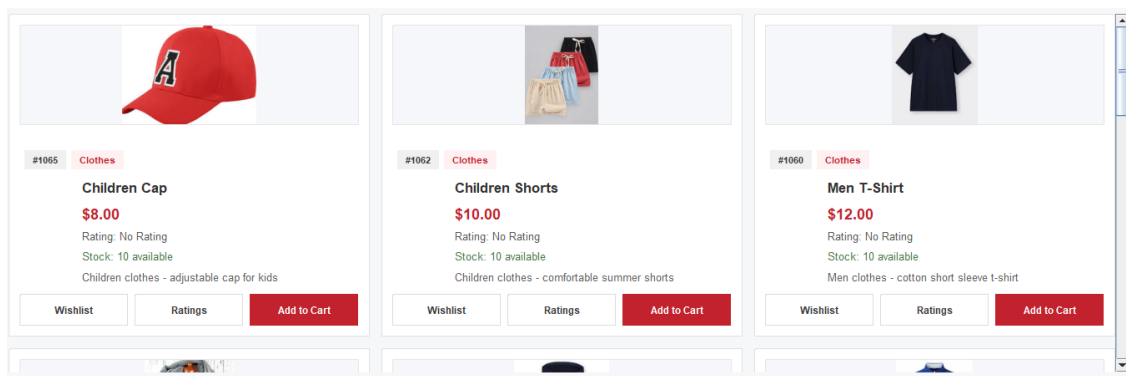


We can Reset Filters by pressing on the Reset Filters Button

Another Thing that we can see the products by clicking on the category we want, Home contains all the products form the store:

Example:

If we choose or pressed on Clothes Category, we will see this:



## For Search feature:

My Wishlist

Wo

Search

sories

Electronics & Arduino

Used Products

Mobile Phones & Tablets

Mobile & Tablet Accessories

Clothes

Perfumes & Watches

Boots & Socks

Home Electronics

Home Acces

## Welcome to MixMart Online Store

MixMart Online Store is a complete shopping platform for electronics, accessories, home products, fashion items, books, toys, and used products.

Customers can browse products, search and filter items, add products to the cart or wishlist, confirm orders, choose flexible delivery, rate products and the website, and submit used items for sale.

Rate Website

### Flexible Delivery Option

Choose from 3 shipping companies to the delivery option that suits you bes

### Easy Payment Methods

Pay comfortably using Cash on Deliver  
Card Payment.

### Our Policies

Click here to View our store rules, deliv  
terms, and payment policies.

#1064

Clothes



**Women Scarf**  
**\$14.00**  
 Rating: No Rating  
 Stock: 10 available  
 Women clothes - soft scarf for winter

Wishlist

Ratings

Add to Cart

#1058

Clothes

SALE



**Women Long Dress**  
**\$45.00 \$40.50 (10% OFF)**  
 Rating: No Rating  
 Stock: 10 available  
 Women clothes - elegant long dress

Wishlist

Ratings

Add to Cart

#1066

Clothes



**Women Abaya**  
**\$50.00**  
 Rating: No Rating  
 Stock: 10 available  
 Women clothes - simple and elegant abaya

Wishlist

Ratings

Add to Cart

## Search with filters in Home:

MixMart Online Store

High quality products • User friendly website • All in one place

Call us: 055-299-0922 | mmladsamds@gmail.com

MixMart Online Store

Electronics • Accessories • Used Products • etc...

My Wishlist

DD

Search

My Account

Orders

Cart

0 Items

Home

Computer Parts

Computer Accessories

Electronics & Arduino

Used Products

Mobile Phones & Tablets

Mobile & Tablet Accessories

Clothes

Perfumes & Watches

Boots & Socks

Home Electronics

Home Accessories

Books

Kids Toys

Garden Accessories

### Welcome to MixMart Online Store

MixMart Online Store is a complete shopping platform for electronics, accessories, home products, fashion items, books, toys, and used products.

Customers can browse products, search and filter items, add products to the cart or wishlist, confirm orders, choose flexible delivery, rate products and the website, and submit used items for sale.

Rate Website

### Flexible Delivery Options

Choose from 3 shipping companies to get the delivery option that suits you best..

### Easy Payment Methods

Pay comfortably using Cash on Delivery or Card Payment.

### Our Policies

Click here to View our store rules, delivery terms, and payment policies.

#1116

Kids Toys



**Stuffed Teddy Bear**  
**\$18.00**  
 Rating: No Rating  
 Stock: 10 available  
 Soft teddy bear toy for children

Wishlist

Ratings

Add to Cart

#1013

Computer Accessories



**Gaming Mouse DeathAdder**  
**\$28.00**  
 Rating: No Rating  
 Stock: 10 available  
 Ryzzen Wired gaming mouse with high DPI

Wishlist

Ratings

Add to Cart

#1001

Computer Parts



**Crucial DDR4 RAM 16GB**  
**\$45.00**  
 Rating: 5.0 / 5  
 Stock: 10 available  
 16GB DDR4 desktop RAM for gaming and office PCs

Wishlist

Ratings

Add to Cart

4 product(s) shown

Checkout

Filters

Price: Low to High

Only discounted (SALE)

Reset Filters

Sell your product

Submit your own item for admin approval.

Show your item

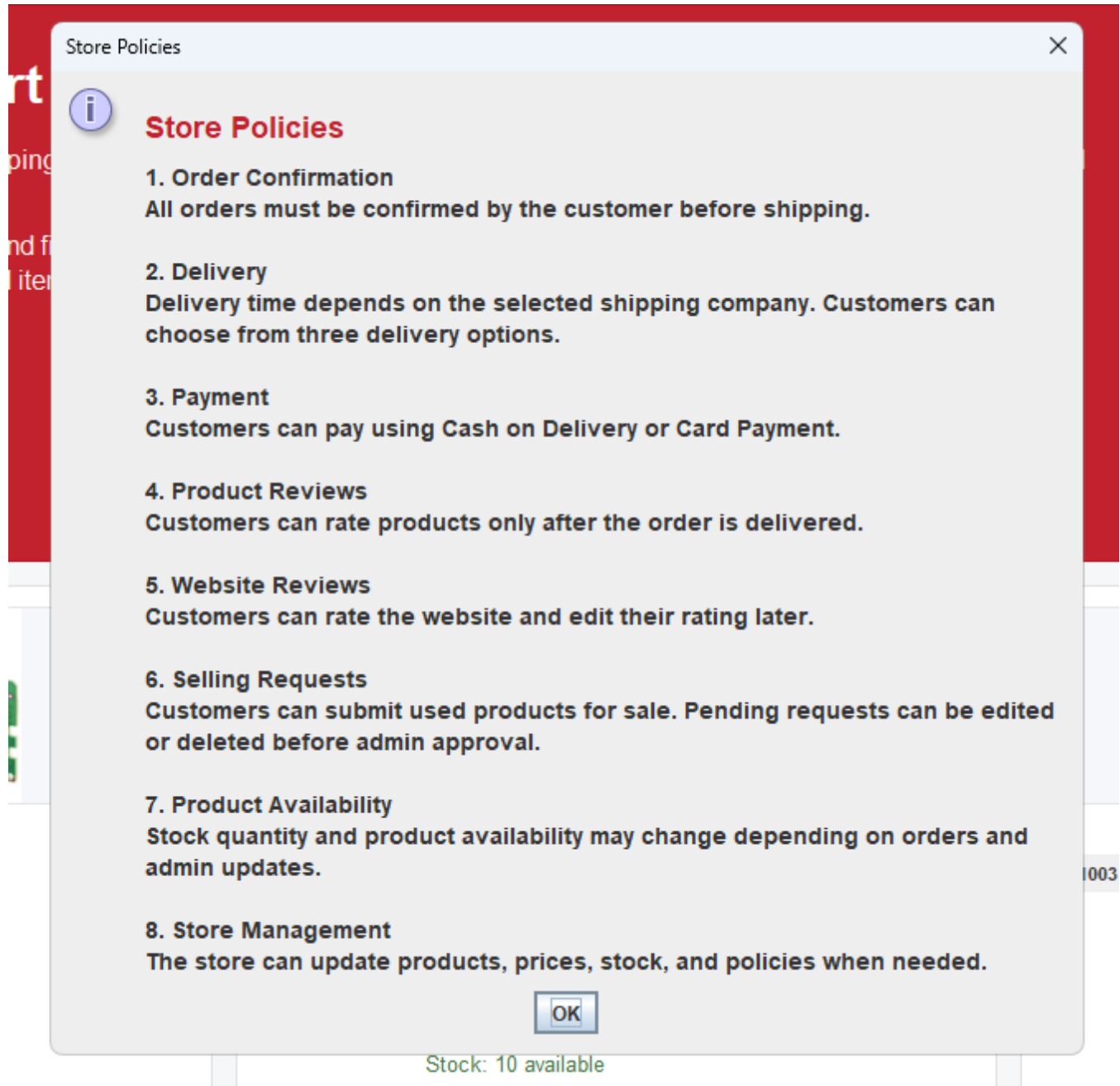
My Selling Requests

Support Chat

Need help? Chat with support.

Open Chat

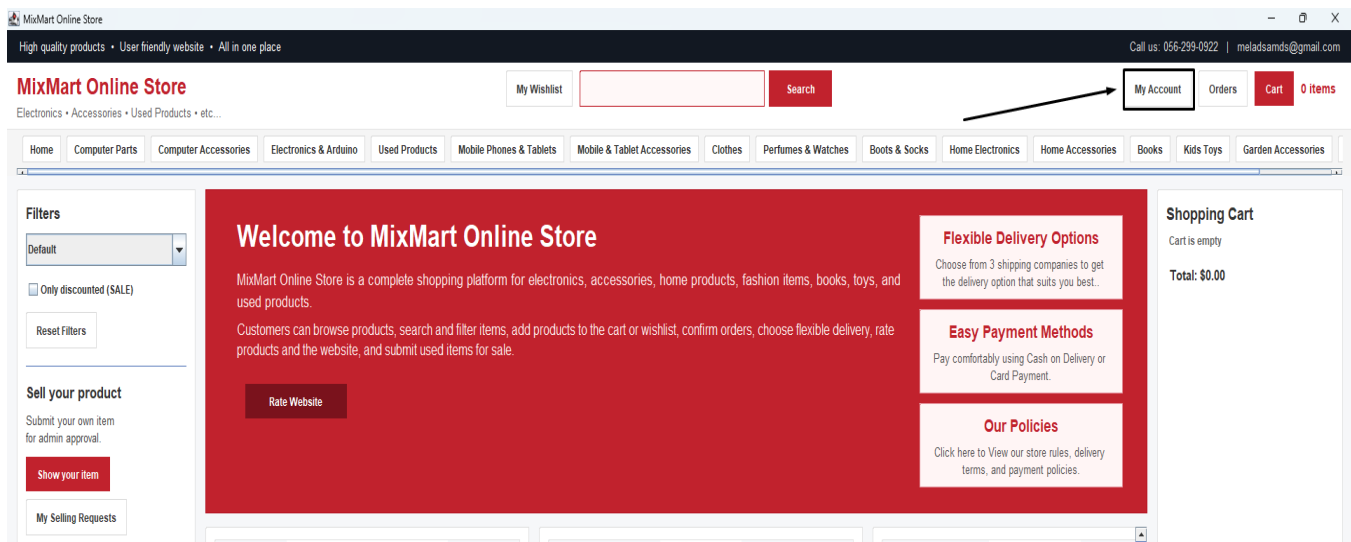
If we press in Our Policies it will open this window for us :



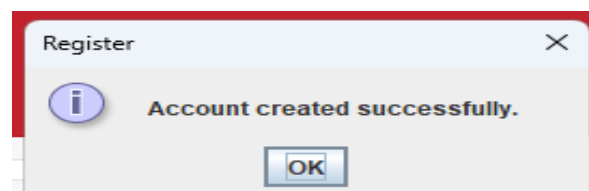
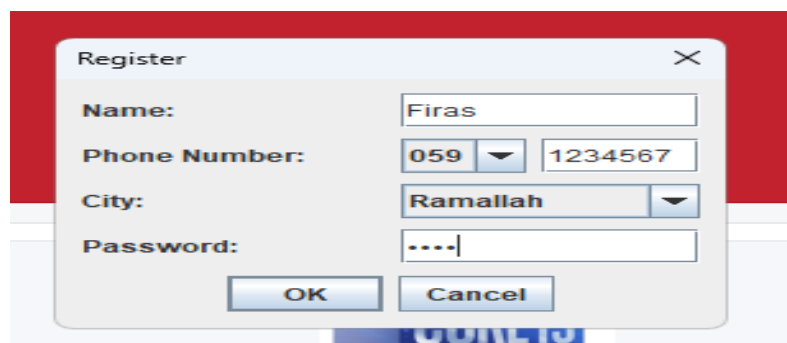
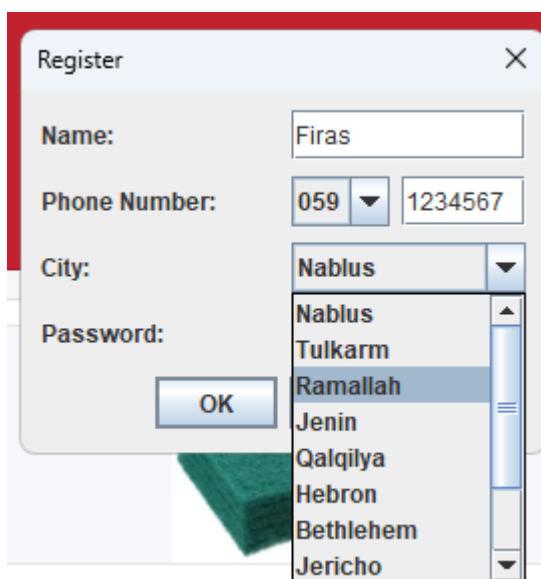
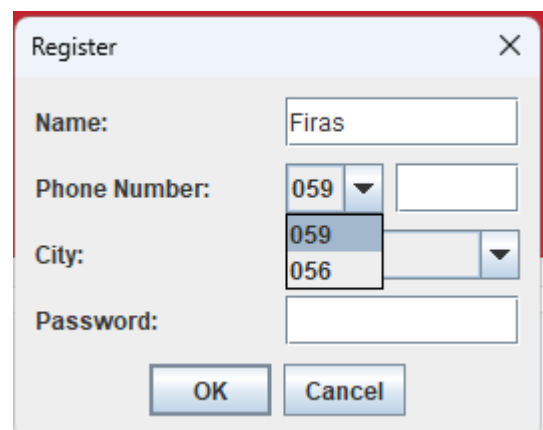
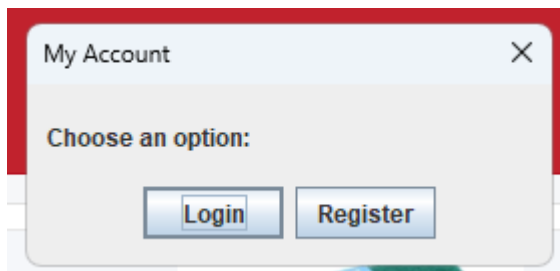
Stock: 10 available

## Registration and Login Form

To get access to everything, we need to have an account by Making new one (Register) or login

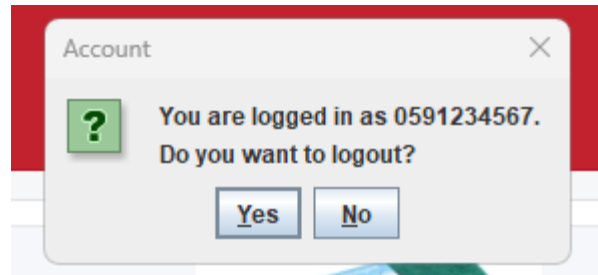
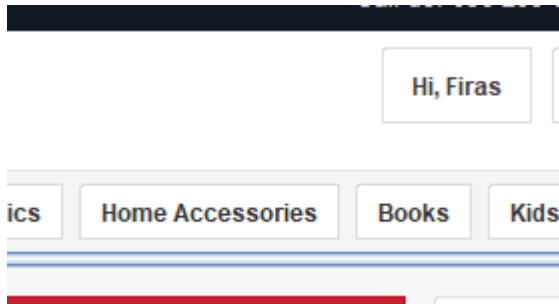


Choose Register:



### To logout:

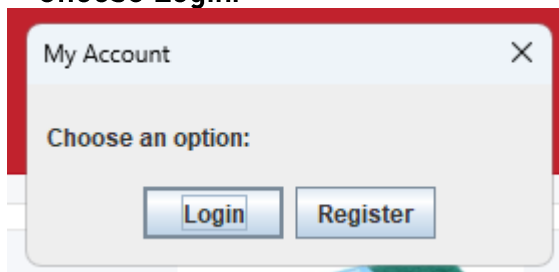
Press on your name and then yes



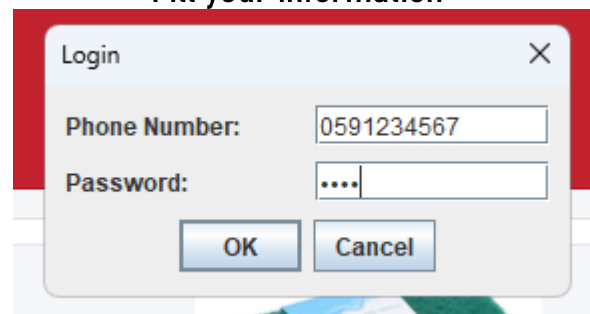
To login again You need to press

**My Account**

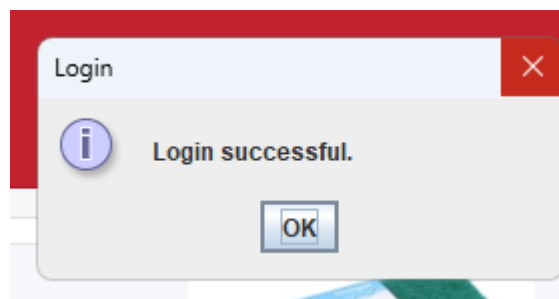
### Choose Login:



### Fill your information



And Now we can use all the fetures:

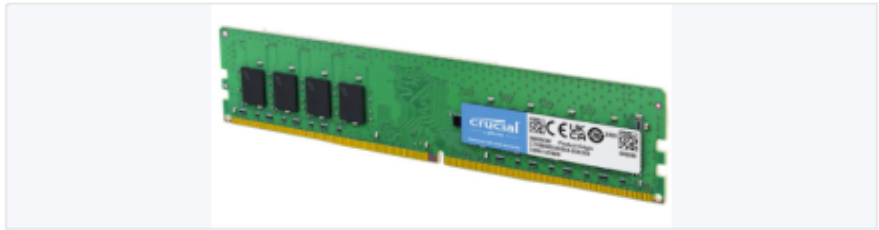


**Note:** The admin is an Regular Customer but has an special GUI that have accses to on the store

we will get to it soon.

Cart, Checkout and Wishlist

Users have Wishlist, for example if we press the Wishlist button this DDR4 Ram will be added to My Wishlist, and i can remove it whenever i wants to.



#1001 Computer Parts

Crucial DDR4 RAM 16GB

\$45.00

Rating: 5.0 / 5

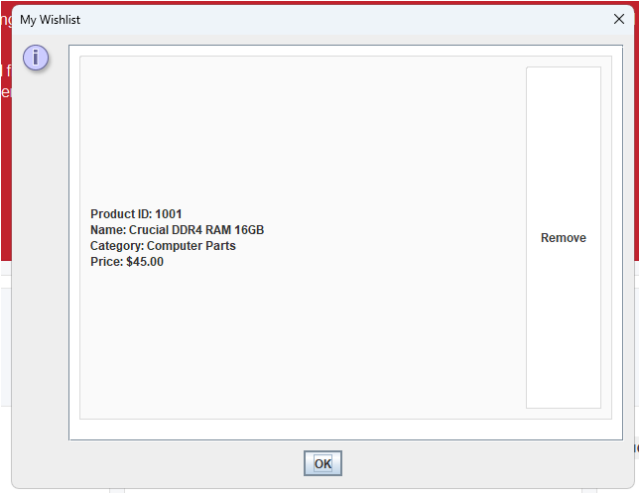
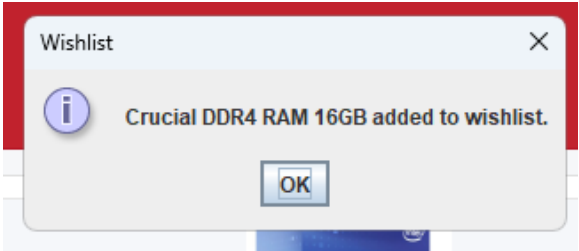
Stock: 10 available

16GB DDR4 desktop RAM for gaming and office PCs

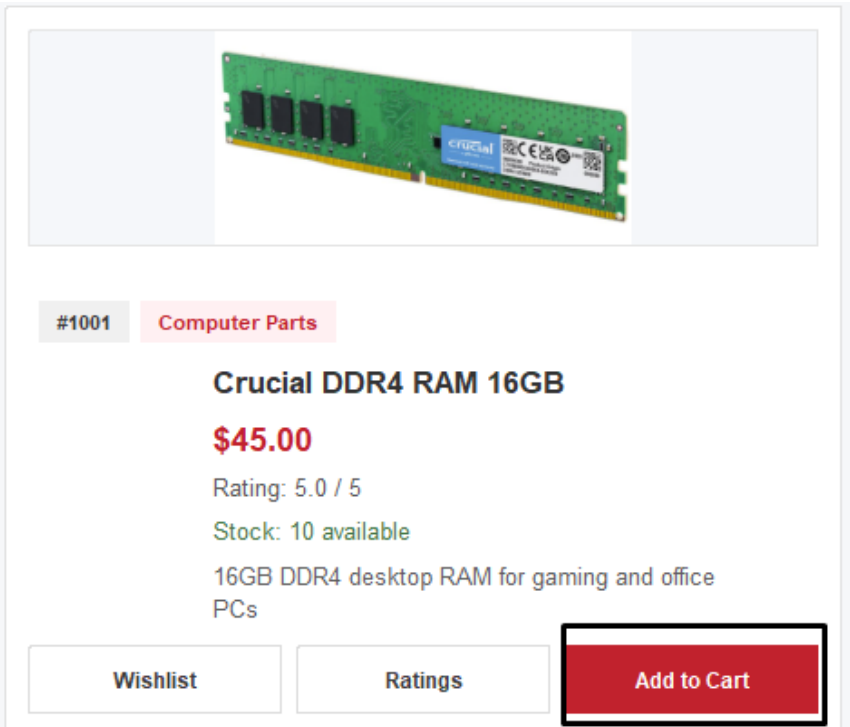
Wishlist

Ratings

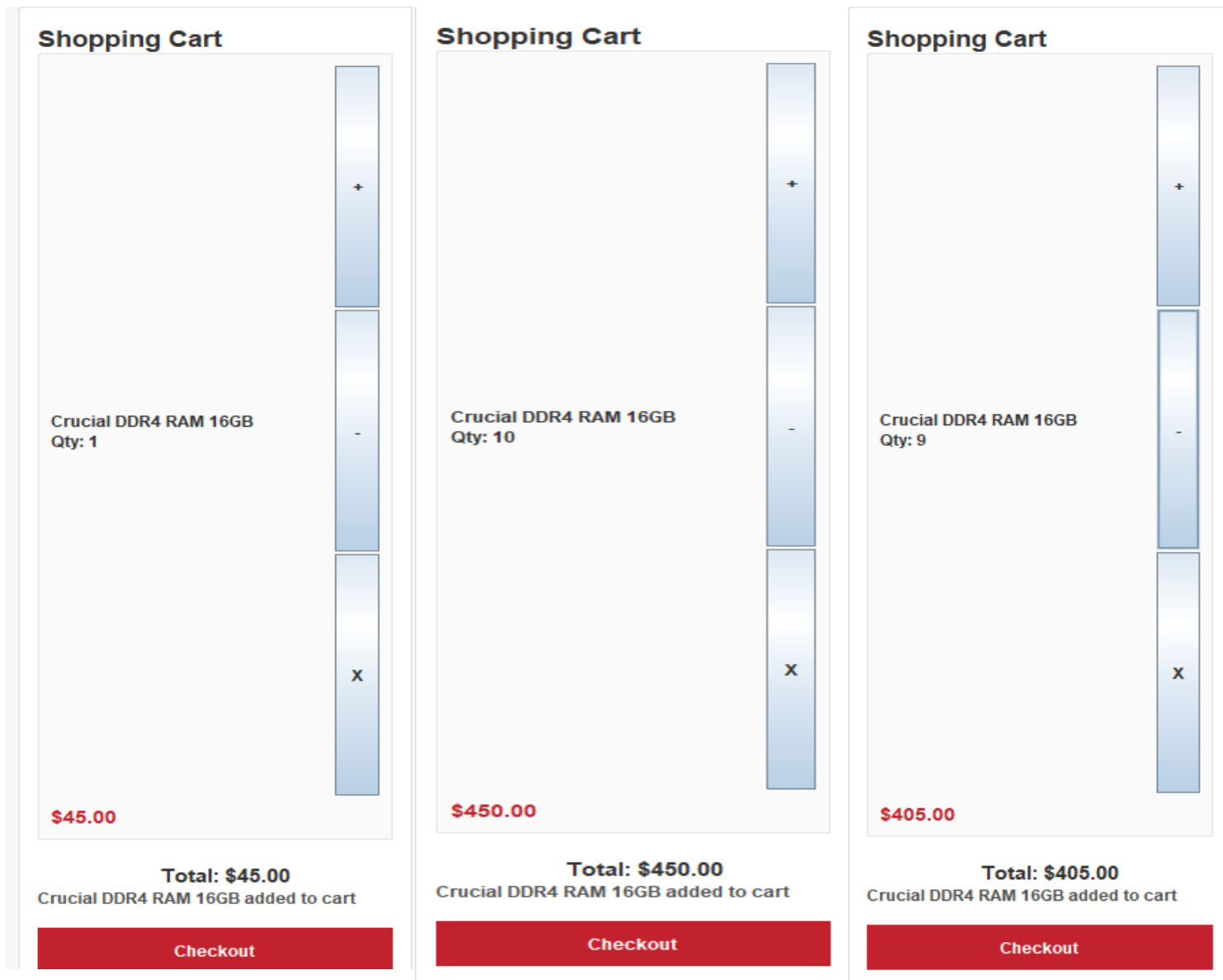
Add to Cart



If we press Add to cart



It will add the Product to the Shopping Cart, and we can Add more by using the + or cancel it from the Shopping cart by using X or use - to make it less, we should not put more than the amount of the product or it will give us error and we can't make it less than 1 we should use X

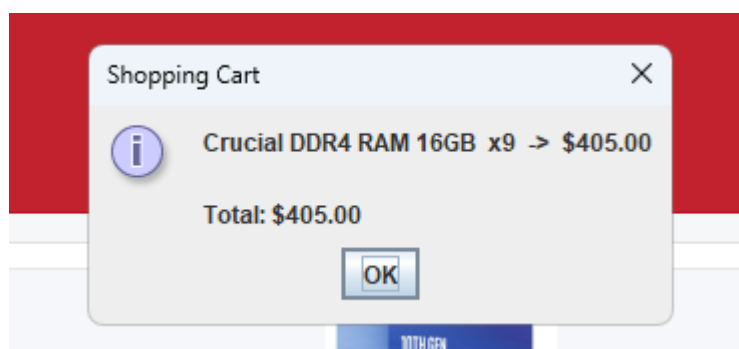


If we press on Cart, Shopping Cart

Cart

9 items

it will give us summary to the items inside of The





If we pressed Checkout, we will need to enter our Receiver information, and it should be true or it will give an error. The Receiver Phone can be any True Phone number, also Payment Method if it is Card or Cash it varies because if you press the cash then it is Cash on Delivery if it is card then you paid before the order is delivered, we will see the details soon

Checkout Information

Receiver Name: Firas

Receiver Phone: 0591234567

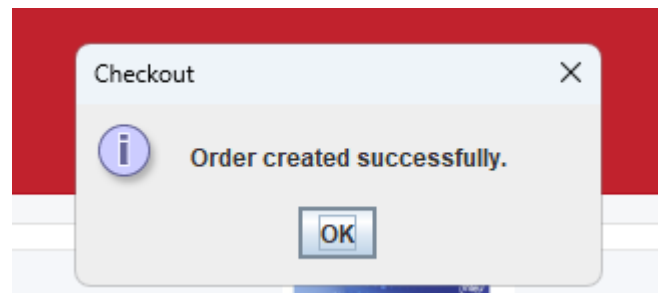
City: Nablus

Street: Najah

Payment Method: Cash

OK

Here I chose Card and then pressed OK:



## Orders, Confirmation and Rating

If we pressed on the **Orders** it will give us all the orders the user did the Checkout to them waiting for user confirmation

My Orders

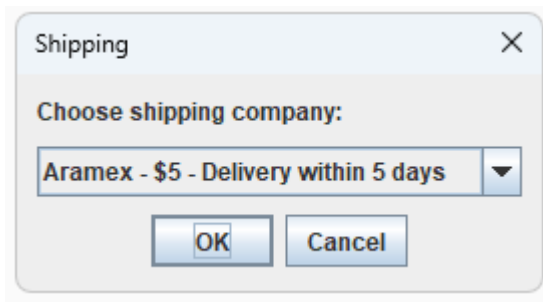
Order ID: 4013  
 Date: 2026-05-16 16:59:36.0  
 Status: Pending  
 Total: \$405.00  
 Receiver: Firas  
 Receiver Phone: 0591234567  
 Address: Nablus, Najah  
 Payment: Card / Pending

Confirm

OK

And if we pressed Confirm it will ask us to enter the Shipping company, and our Store have Flexible Delivery Options.

Note: 1day in the store equals 1min on the clock

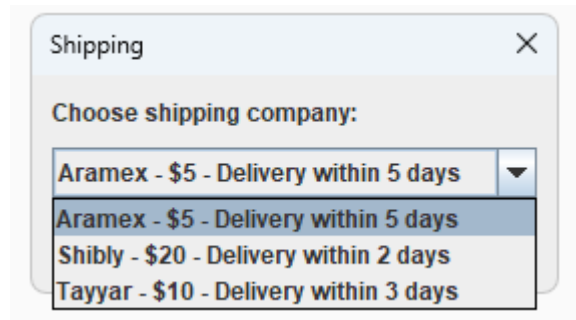


Shipping

Choose shipping company:

Aramex - \$5 - Delivery within 5 days

OK Cancel



Shipping

Choose shipping company:

Aramex - \$5 - Delivery within 5 days

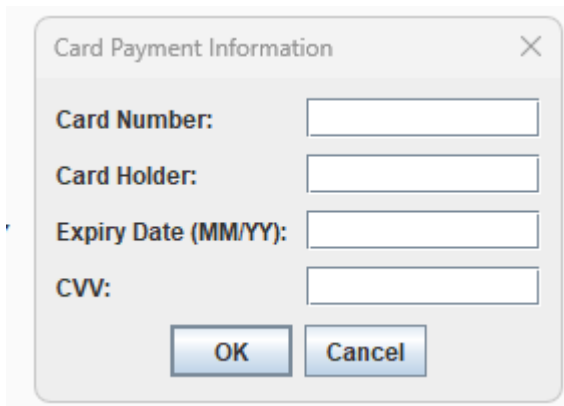
Aramex - \$5 - Delivery within 5 days

Shibly - \$20 - Delivery within 2 days

Tayyar - \$10 - Delivery within 3 days

and because we choose card it will give us this Form after pressing ok:

and also if we entered an information that are wrong or break rules it will give you an error:



Card Payment Information

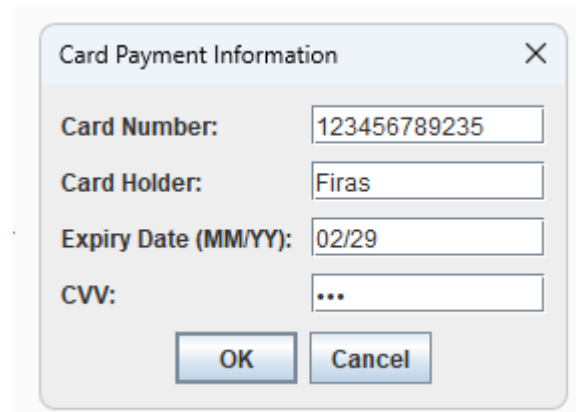
Card Number:

Card Holder:

Expiry Date (MM/YY):

CVV:

OK Cancel



Card Payment Information

Card Number: 123456789235

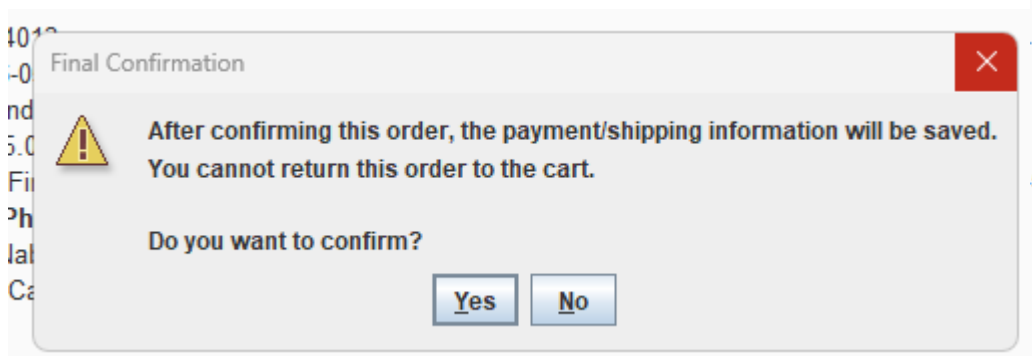
Card Holder: Firas

Expiry Date (MM/YY): 02/29

CVV: ...

OK Cancel

To make it easier for the SQL we put No-return policy, and we ask for Confirmation again:

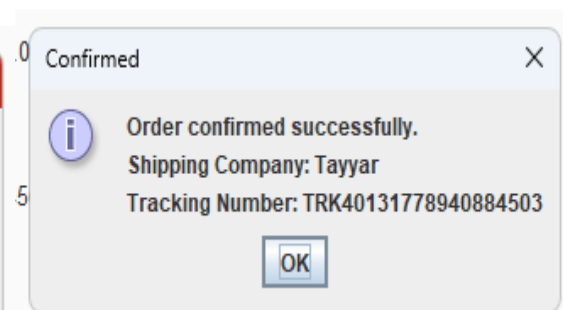


Final Confirmation

After confirming this order, the payment/shipping information will be saved.  
You cannot return this order to the cart.

Do you want to confirm?

Yes No

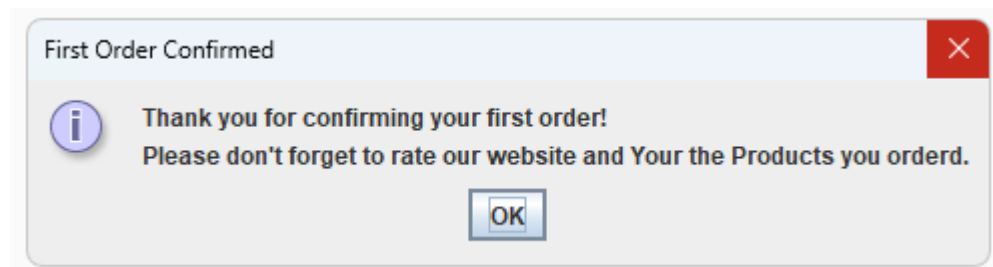


Confirmed

Order confirmed successfully.  
Shipping Company: Tayyar  
Tracking Number: TRK40131778940884503

OK

After Pressing OK



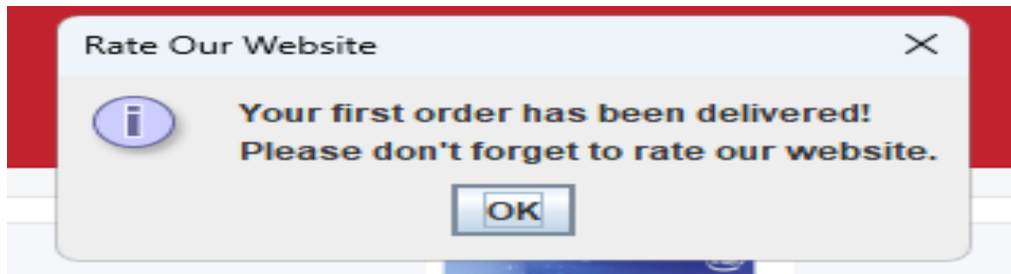
First Order Confirmed

Thank you for confirming your first order!  
Please don't forget to rate our website and Your the Products you orderd.

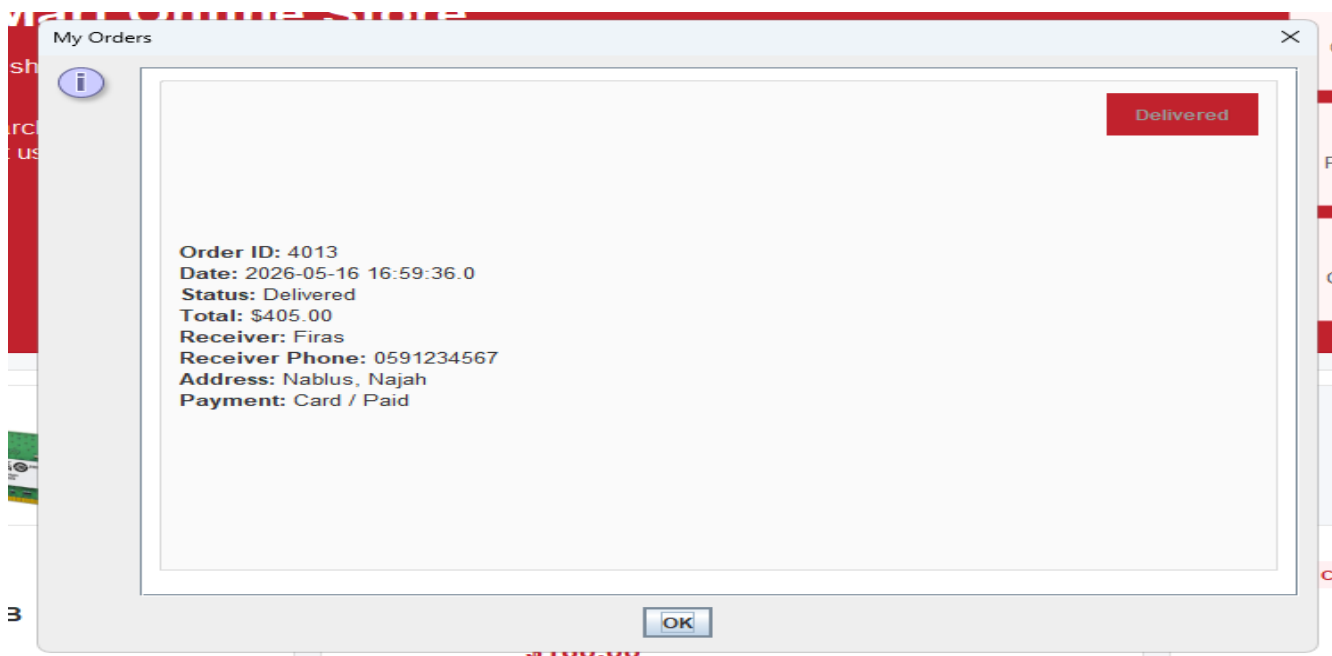
OK

Because we chose Tayyar, the order will take around 3 minutes in real life to be delivered. However, in the system, the order status will be considered as "Shipped". After that, a rating message will appear because it is the user's first order. If the user does not rate the website or any of the purchased products after the order is shipped, the system will send a reminder notification. This reminder will only appear for the first order.

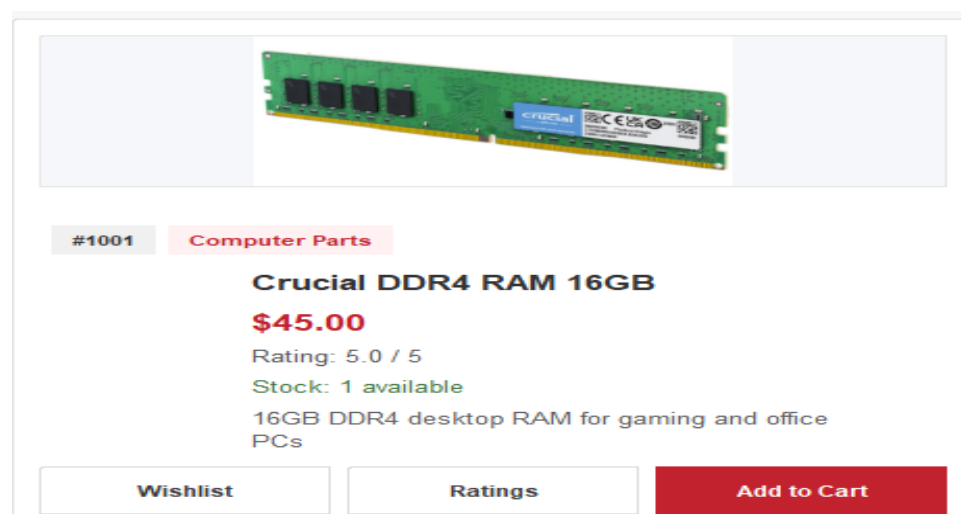
After 3 min if we pressed on Orders and it will give us this, because we didn't rate



And our order is been delivered:



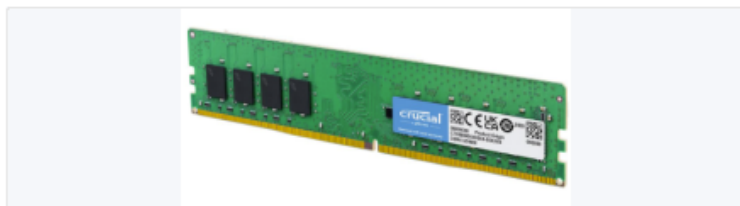
and the DDR4 Ram got 1 available:



Users can only rate and review products that have been shipped or delivered. They can also view existing product ratings and reviews, if available. In addition, users can rate the website after their first order has been shipped or delivered.

Users are allowed to edit their ratings and reviews later if they want. Their previous review and rating will be saved in the system so they can update them in the future.

### Products:



#1001

Computer Parts

#### Crucial DDR4 RAM 16GB

**\$45.00**

Rating: 5.0 / 5

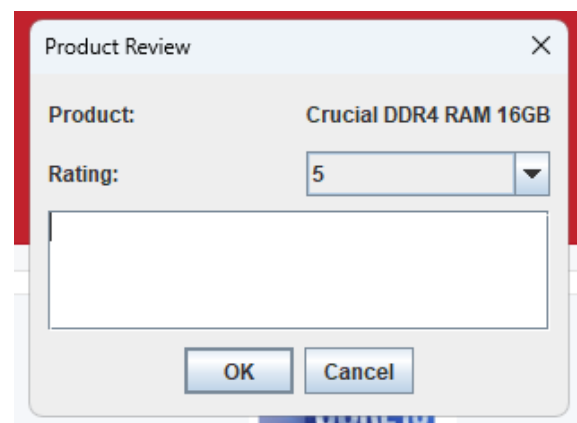
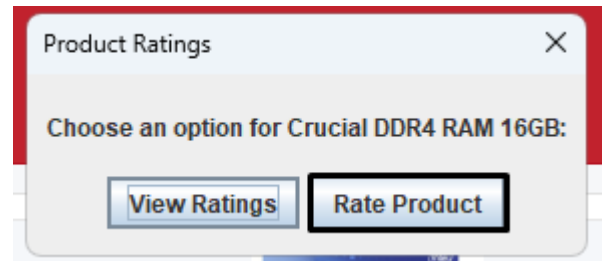
Stock: 1 available

16GB DDR4 desktop RAM for gaming and office PCs

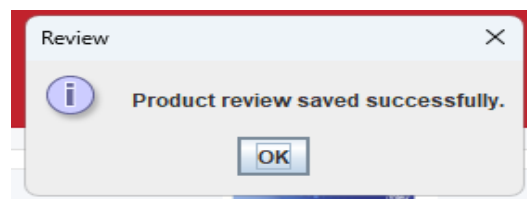
Wishlist

Ratings

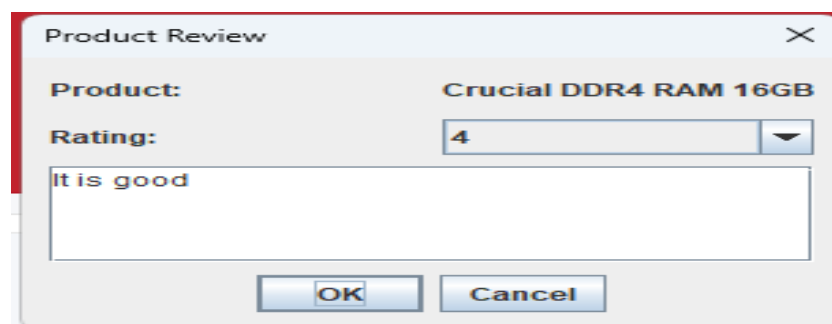
Add to Cart



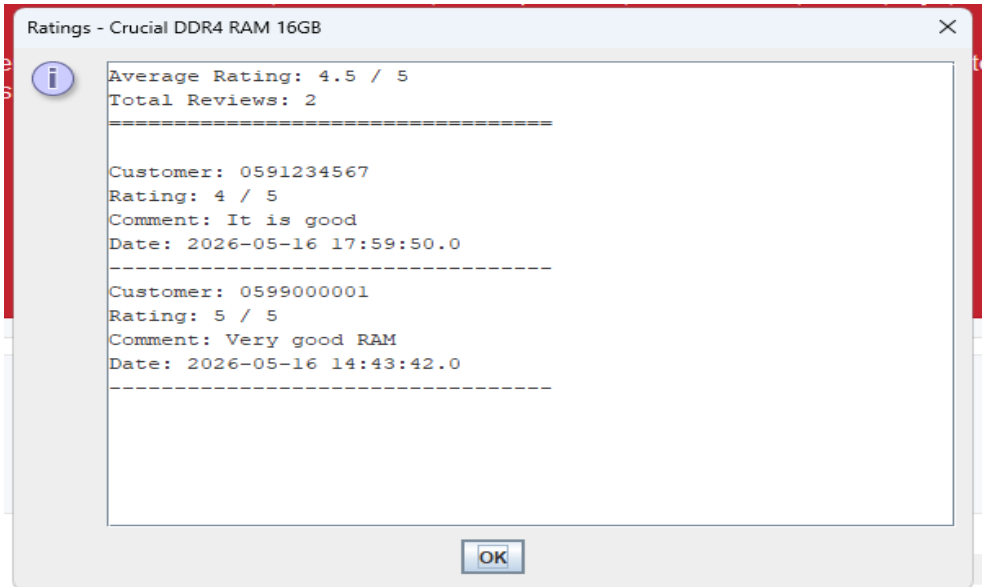
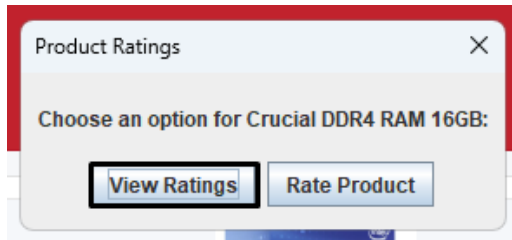
We comment "It is good" then pressed OK



Then if we went back again to the Rate Product:



if we pressed on View Ratings:



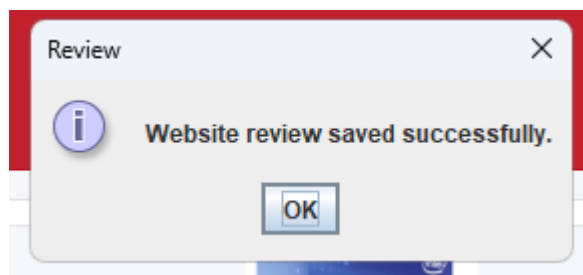
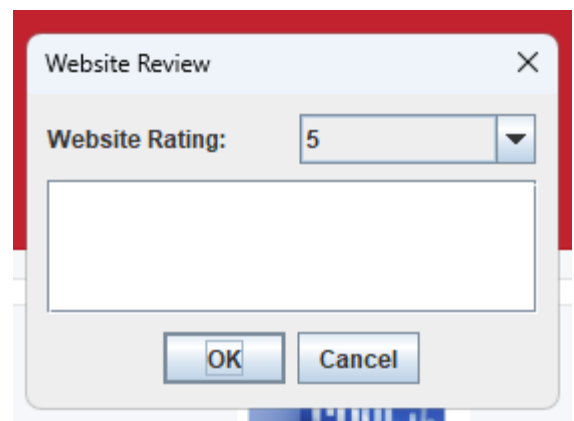
Website:

same as the product

MixMart Online Store is a complete shopping used products.

Customers can browse products, search and filter products and the website, and submit used items.

Rate Website



## Selling your product User side

### Sell your product

Submit your own item  
for admin approval.

Show your item

My Selling Requests

If the user pressed on sell your product:

**Selling Request**

Customer Phone: 0591234567

Product Name:

Product Price:

Product Image:  No image selected

And it is up to the user to fill what information he wants because the admin will later accept this product in his store or not and used product are considered as 1 always and used objects.

After filling the information:

**Selling Request**

Customer Phone: 0591234567

Product Name: i5CPU

Product Price: 123

Product Image:  png-clipart-intel...

**Request Submitted**

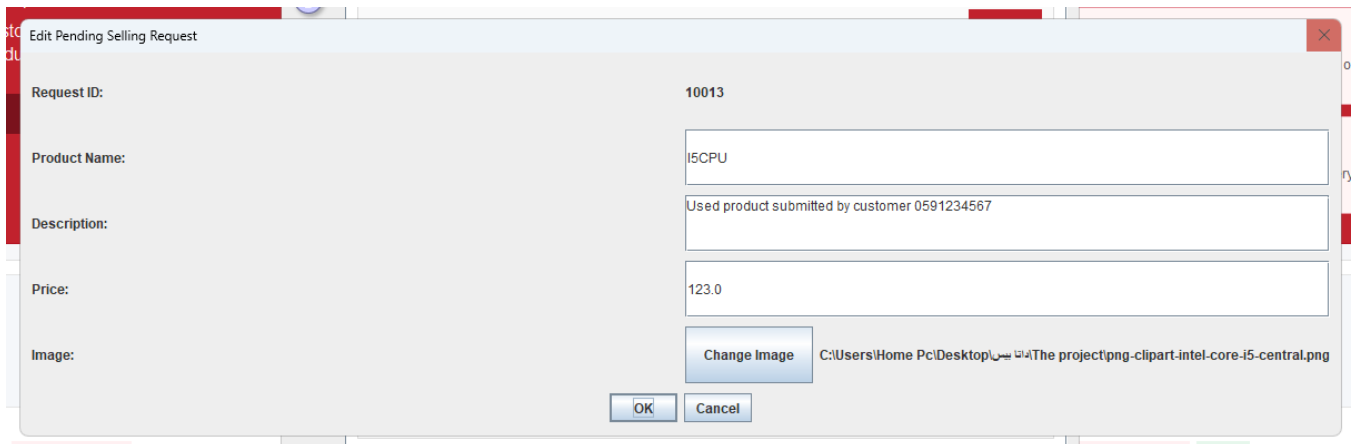
Selling request submitted successfully.  
Your request is now pending admin approval.

After pressing OK and  
then press on My selling  
request:

**My Selling Requests**

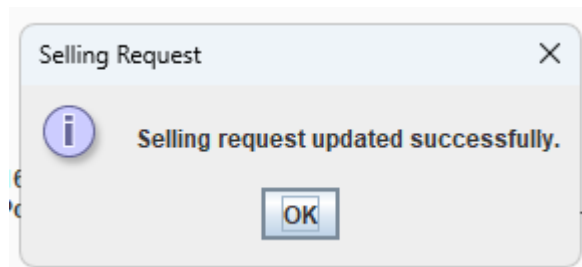
Request ID: 10013  
Product: i5CPU  
Price: \$123.00  
Status: Pending  
Submitted At: 2026-05-16 19:17:31.0  
Image: C:\Users\Home Pci\Desktop\The project\png-clipart-intel-core-i5-central.png

**And while the selling Request is Pending the User/seller can still edit it or delete it:**

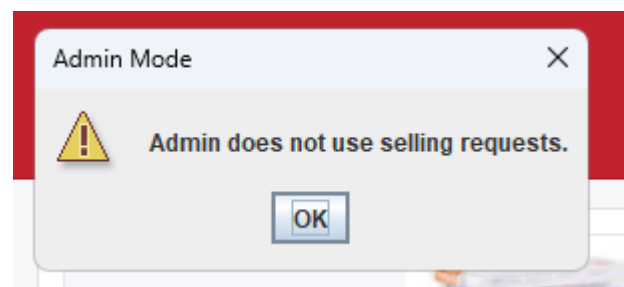
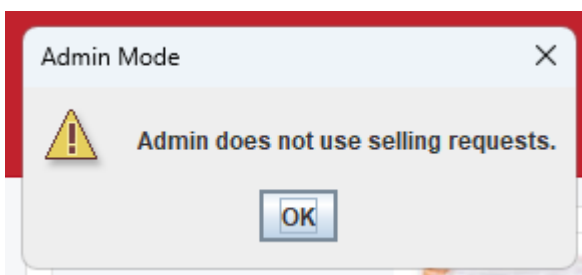


The screenshot shows a dialog box titled "Edit Pending Selling Request". It contains the following fields and controls:

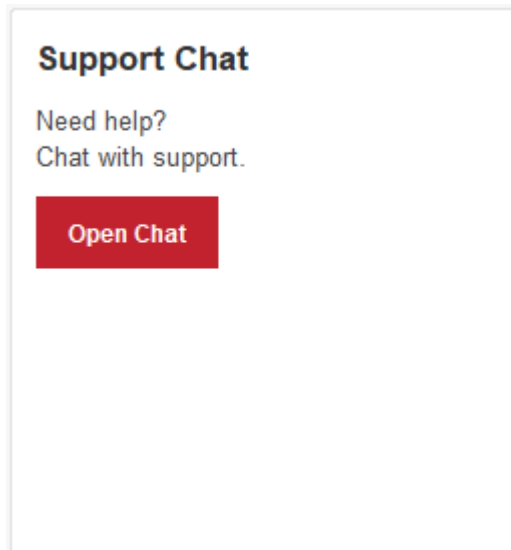
- Request ID:** 10013
- Product Name:** i5CPU
- Description:** Used product submitted by customer 0591234567
- Price:** 123.0
- Image:** A "Change Image" button and a file path: C:\Users\Home Pc\Desktop\بين الماتو\The project\png-clipart-intel-core-i5-central.png
- Buttons:** "OK" and "Cancel" at the bottom.



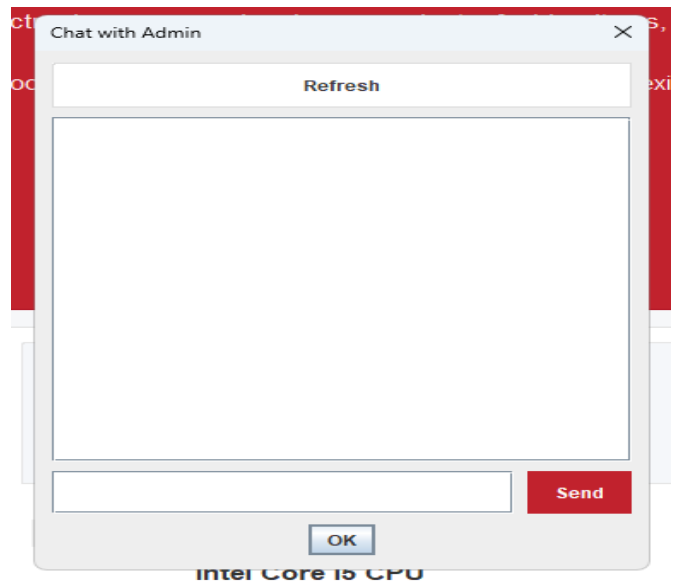
**We will discuss how does the admin accept it when we get to the admin panel**



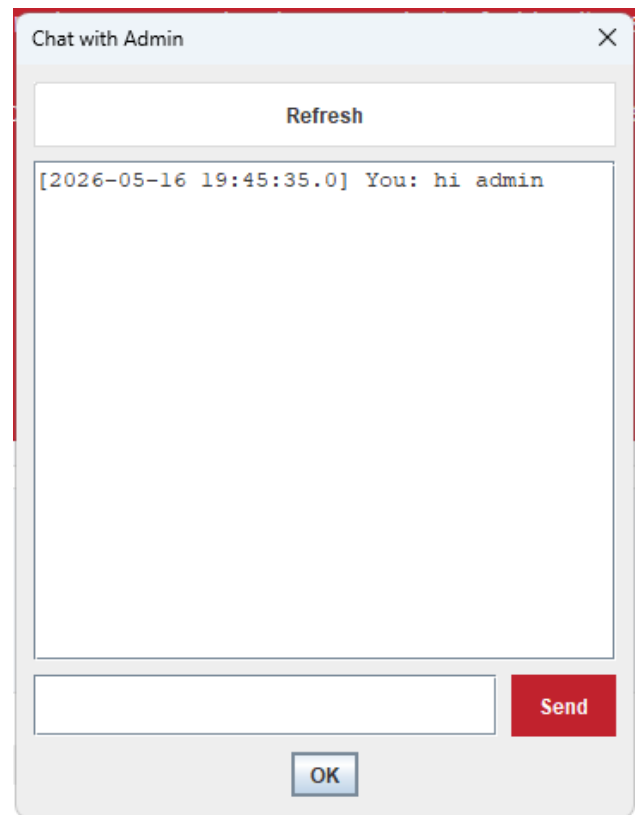
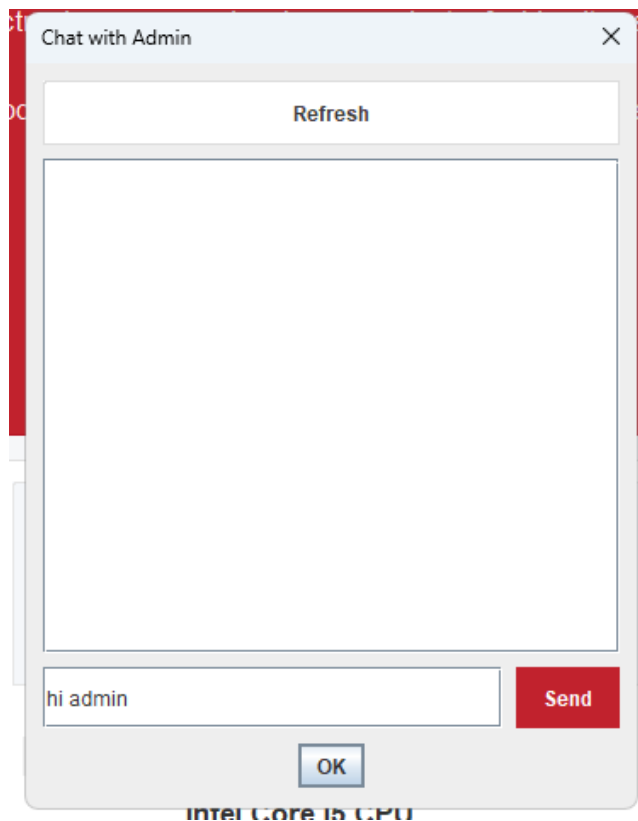
## Support Chat User side



Users can use this feature to talk with admin, as well as the admin to the users,  
When user click on Open Chat:



He can write any thing to the admin and then click on send not OK



**When we get to the admin panel we will discuss the open chat for admin**



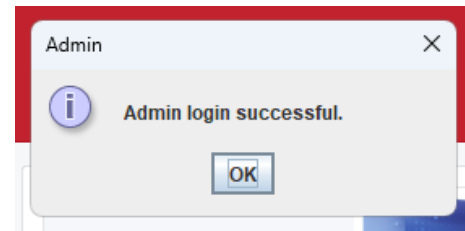
## Admin Panel

Now we will explain the admin panel and what does admin have access to:

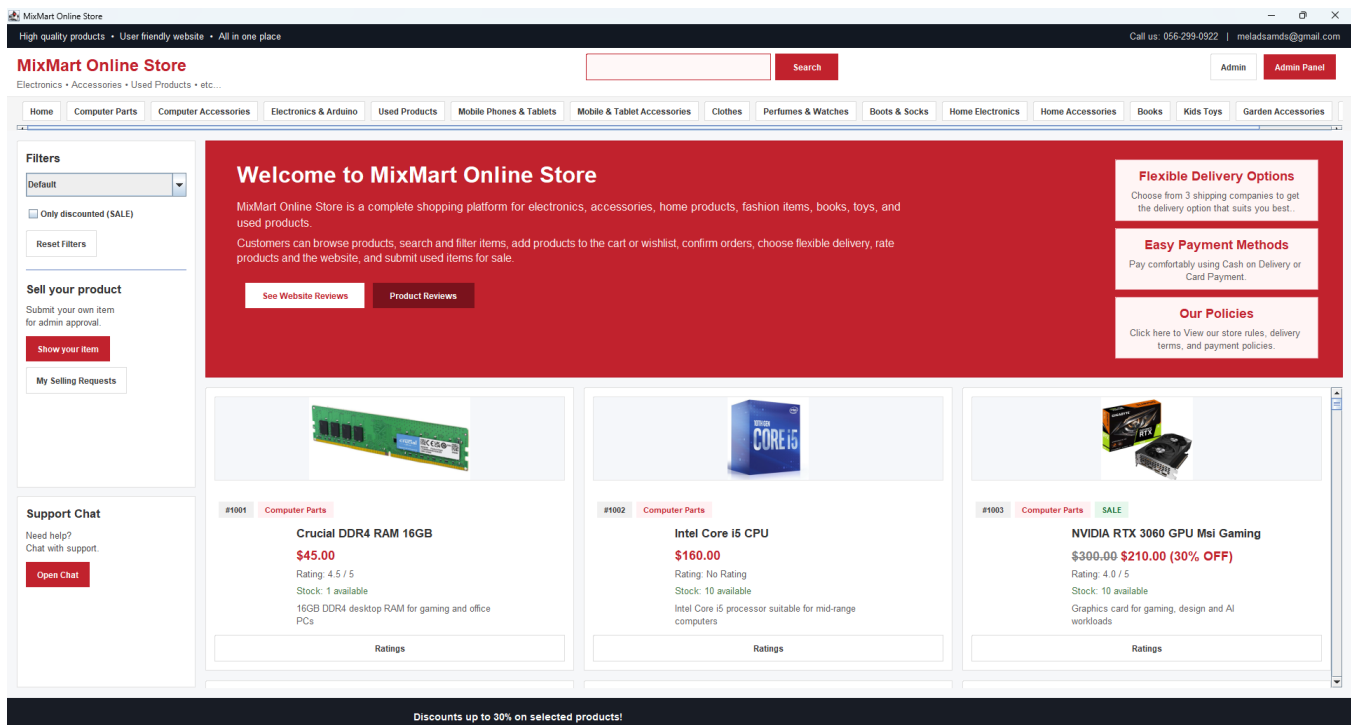
First we login using admins account from the database:

**0593089647: Phone Number**

**3089: Password**



Admin home page does not include (orders, cart, Shopping cart, Rate Website, Add to cart, Wishlist, My Wishlist), so his homepage remain features same as the regular customer



The admin cannot use the “Sell Your Product” feature because it is designed only for regular users. I did not remove the feature from the GUI to avoid affecting the interface design. If the admin tries to access it, the system will display a warning message.

## Lets Start with Admin Ratings:

#1003

Computer Parts

SALE

NVIDIA RTX 3060 GPU Msi Gaming

~~\$300.00~~
**\$210.00 (30% OFF)**

Rating: 4.0 / 5

Stock: 10 available

Graphics card for gaming, design and AI workloads

Ratings

When we press the rating of the as the regular customer sees

Ratings - NVIDIA RTX 3060 GPU Msi Gaming

i

Average Rating: 4.0 / 5

Total Reviews: 1

=====

Customer: 0599000004

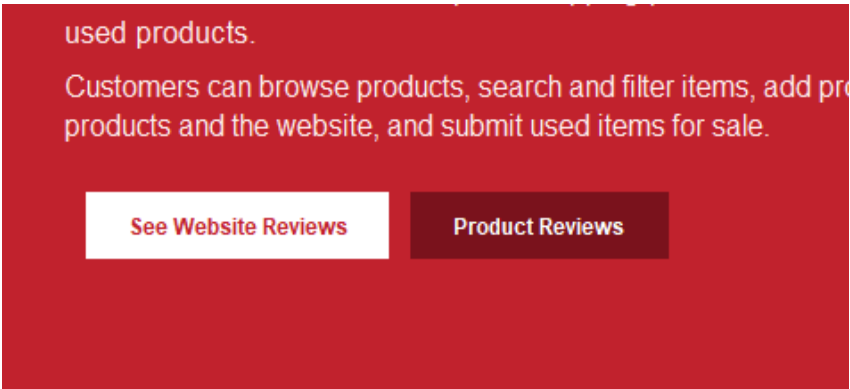
Rating: 4 / 5

Comment: Good GPU for gaming

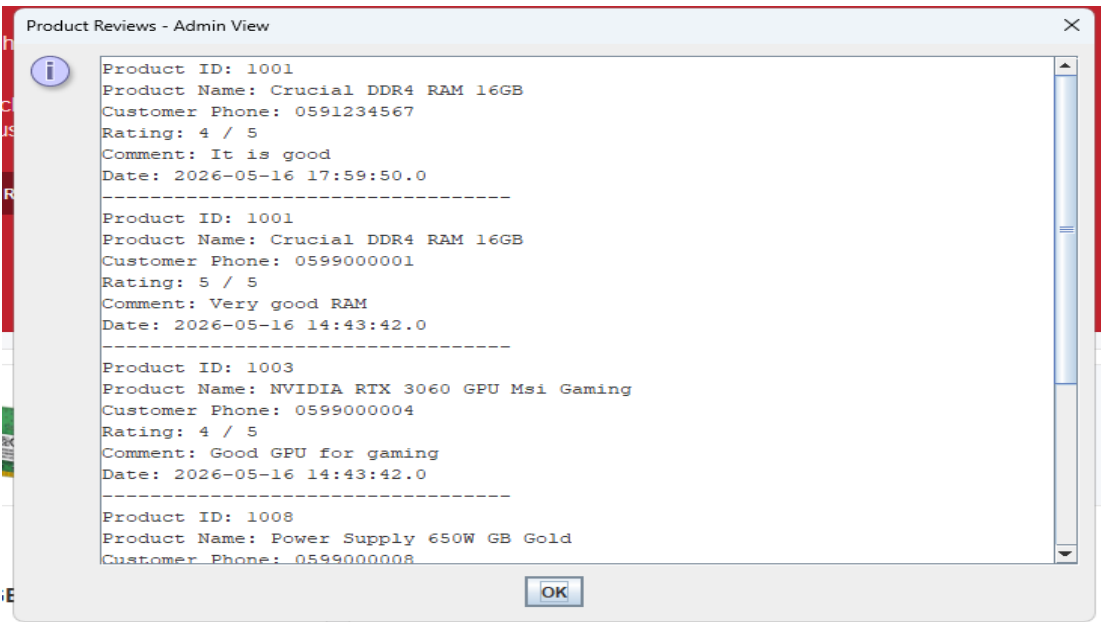
Date: 2026-05-16 14:43:42.0

-----

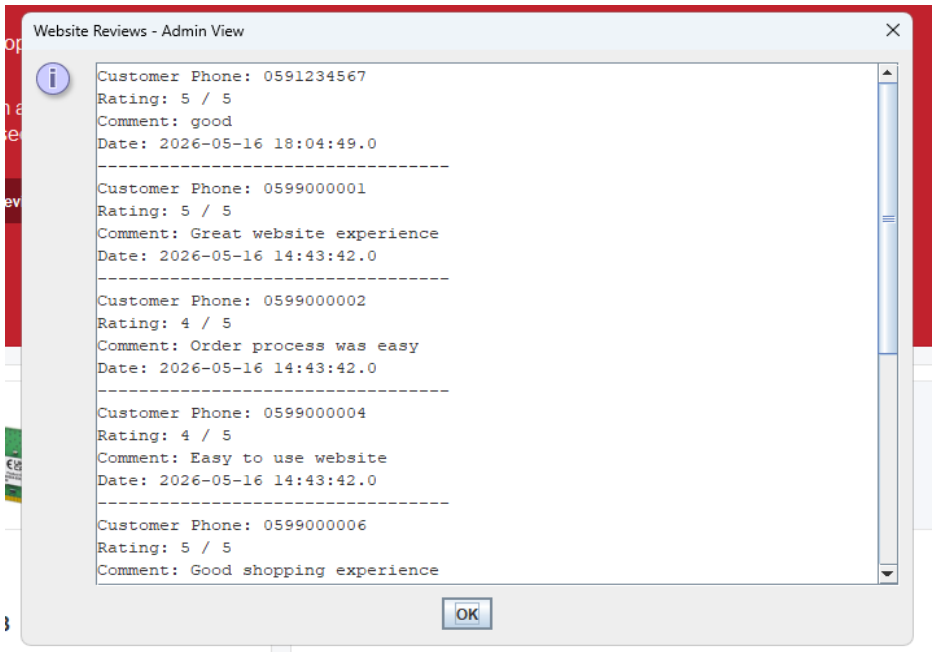
Pressing any of those will open a window to all the Ratings:



Products:

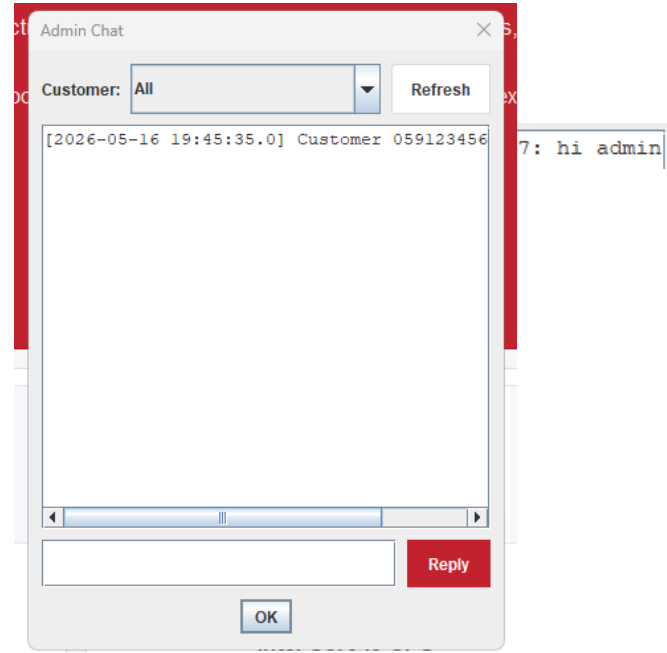


Website:

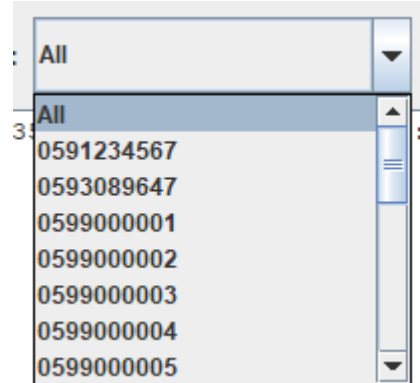


## Support Chat for admin:

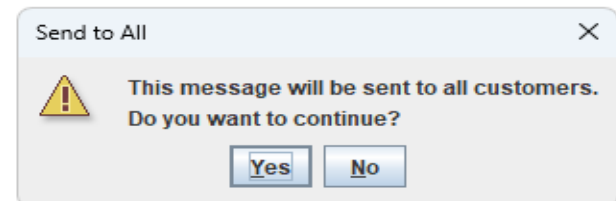
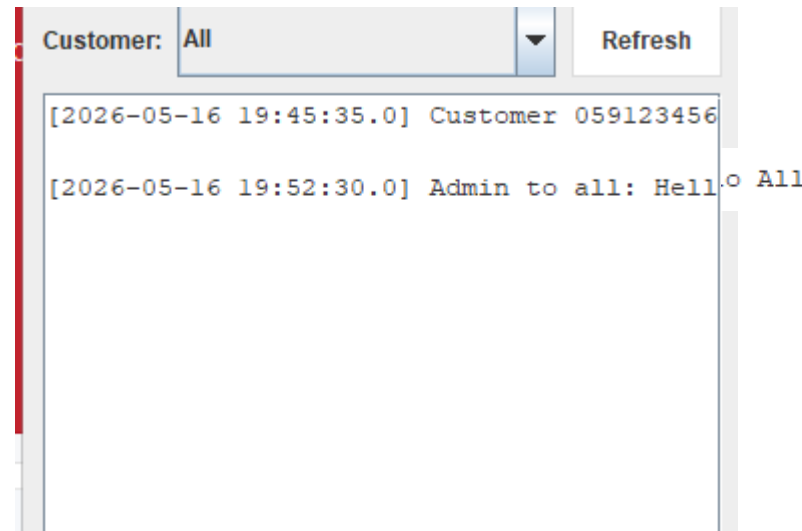
When admin click on the open chat:



admin can sent a message to everyone by All, and sent separately:

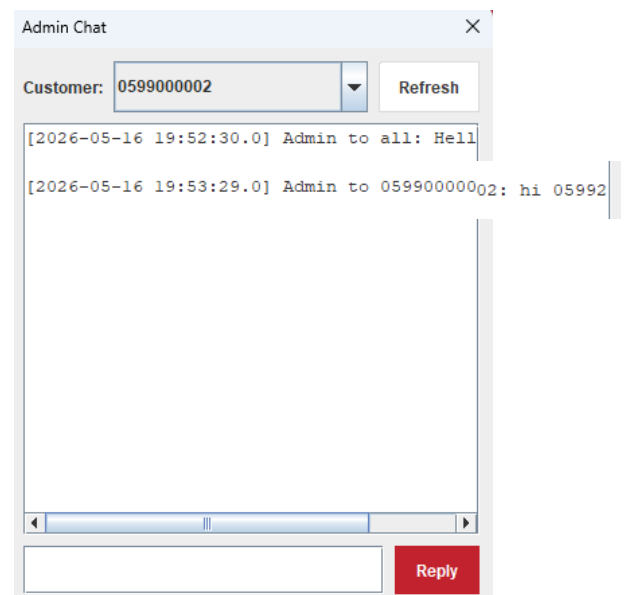


But when he tries to sent All Message it gives him a warning



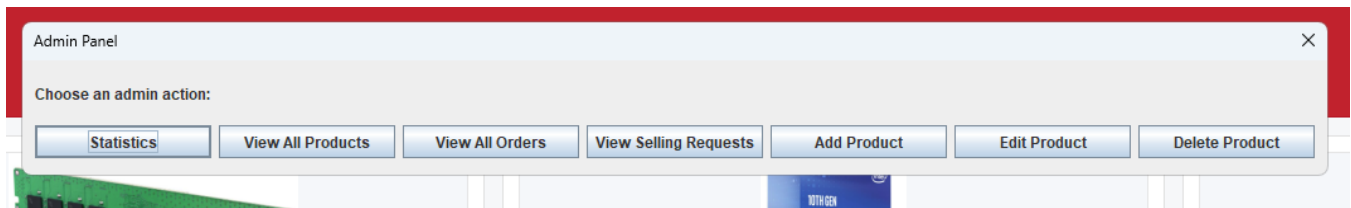
And can see the messages from each Customer separated:

The benefit of the Support Chat is when Saller wants to edit or delete or any help that can't be done in seconds so he messages the owner and they help him

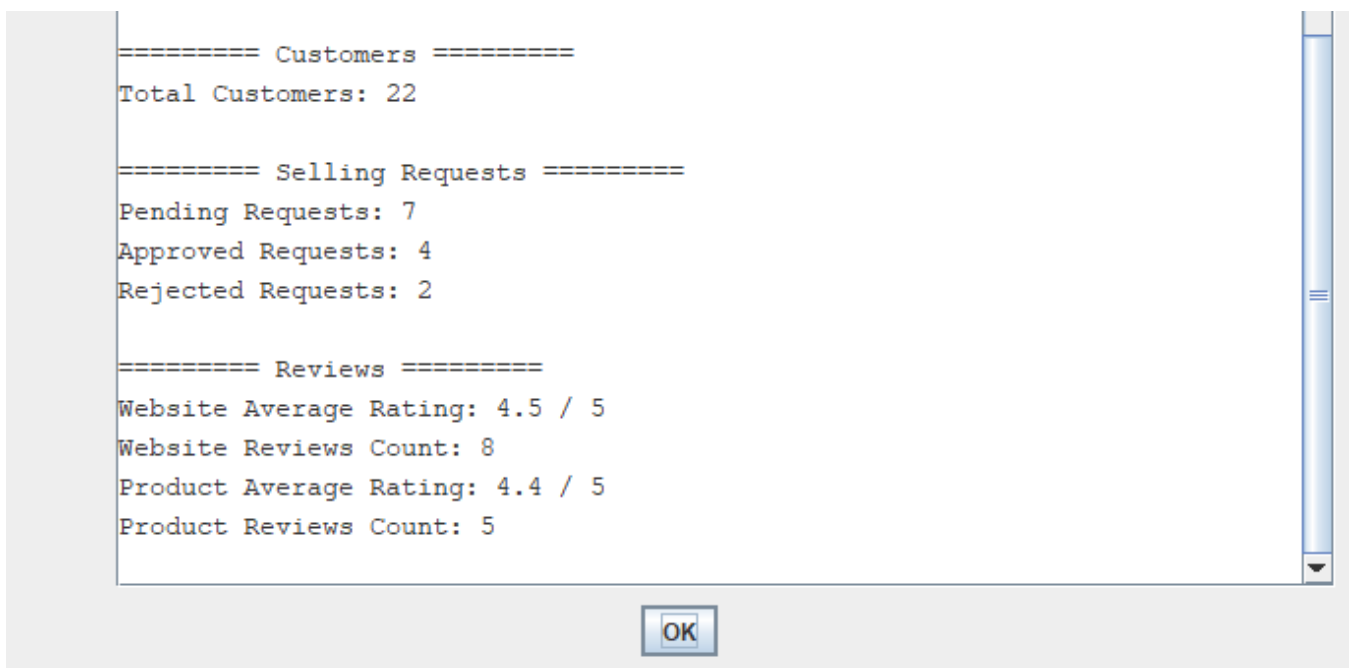
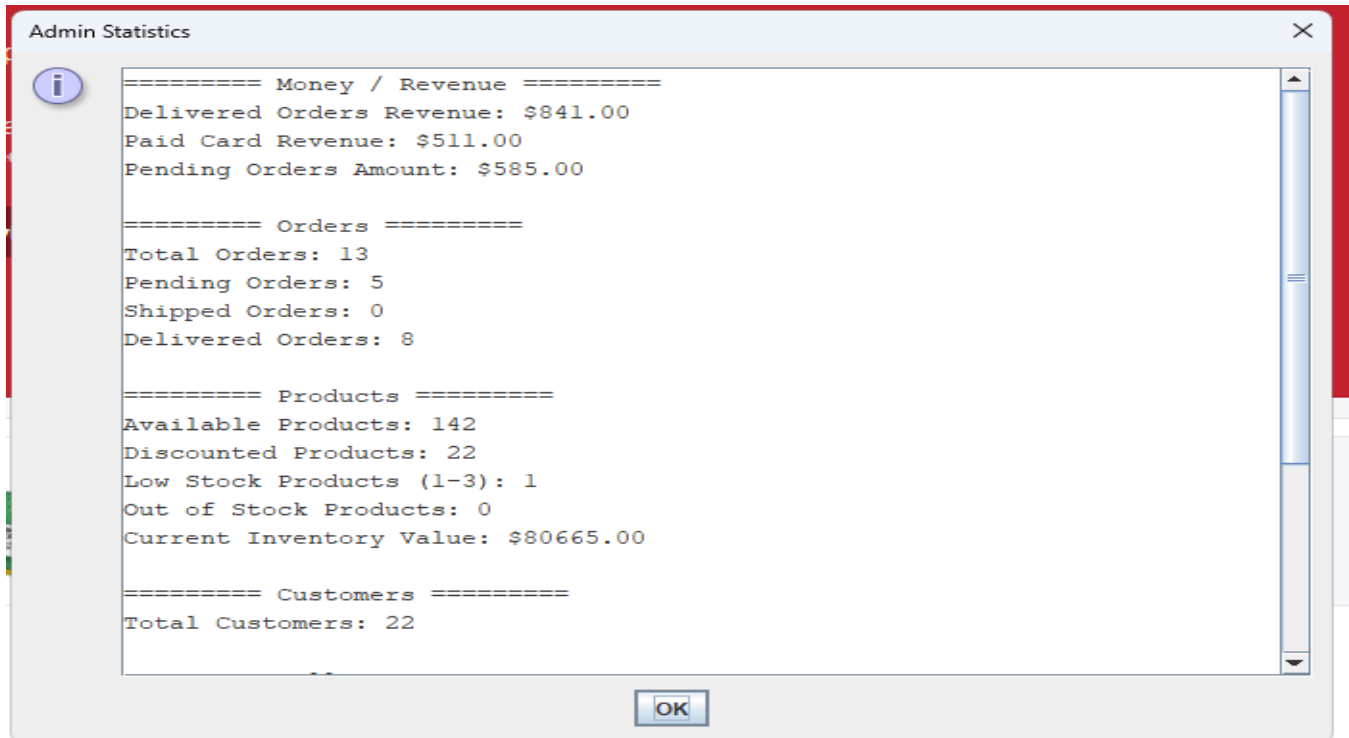


## Now the Admin Panel:

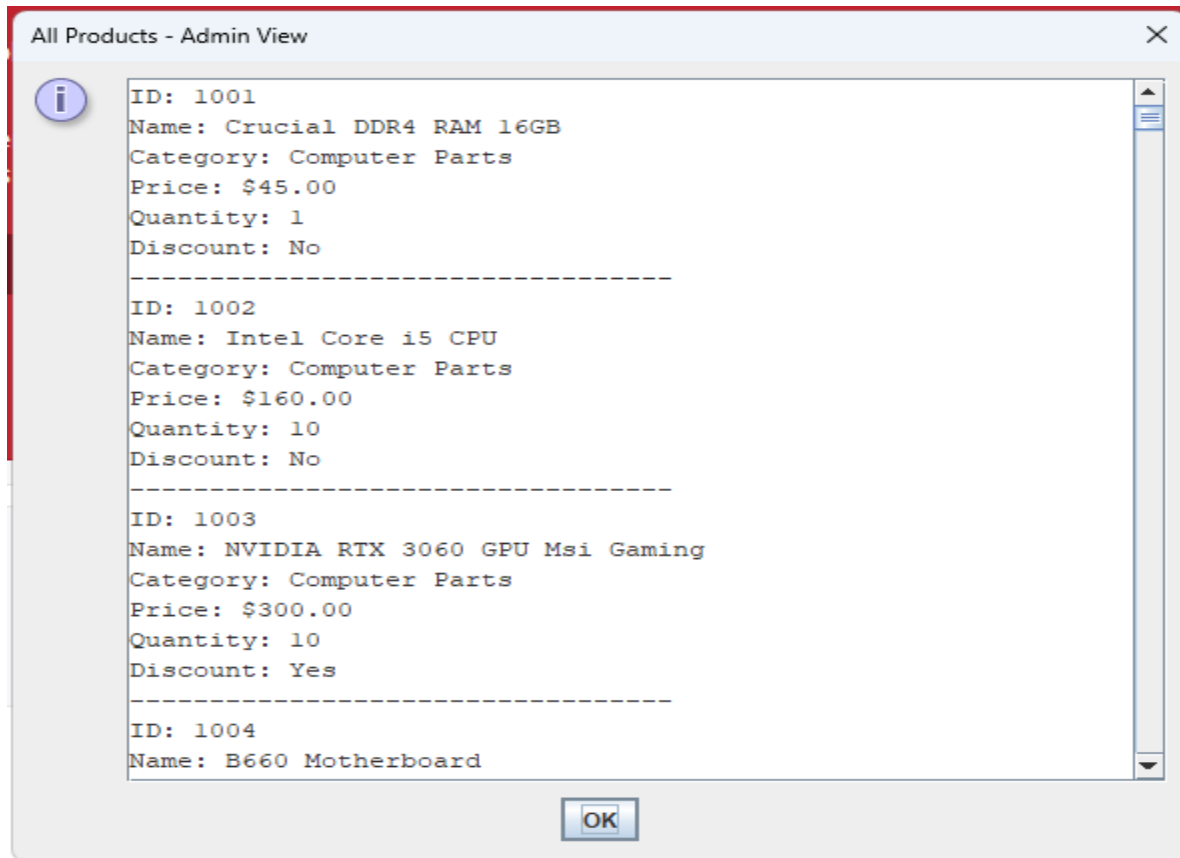
## Admin Panel



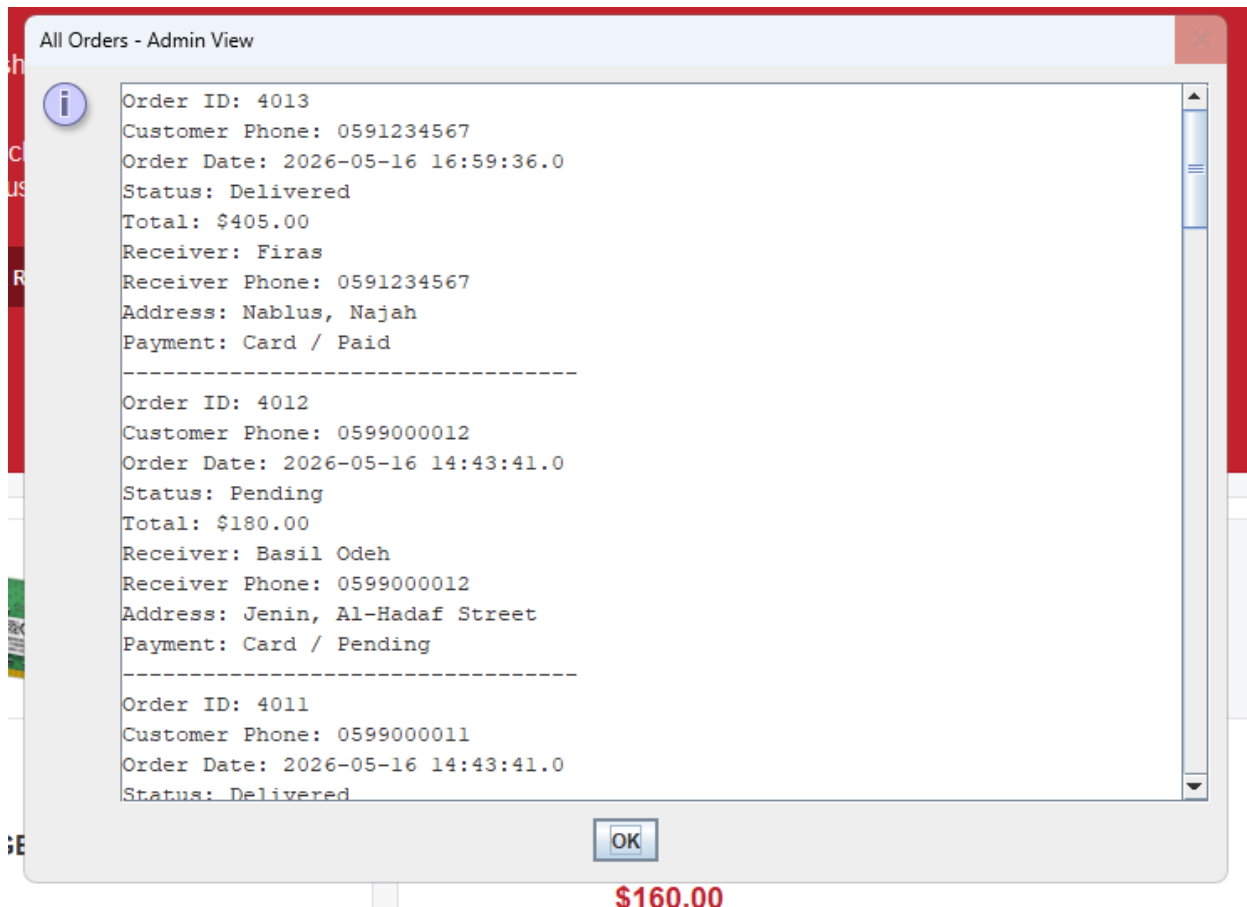
## Statistics:

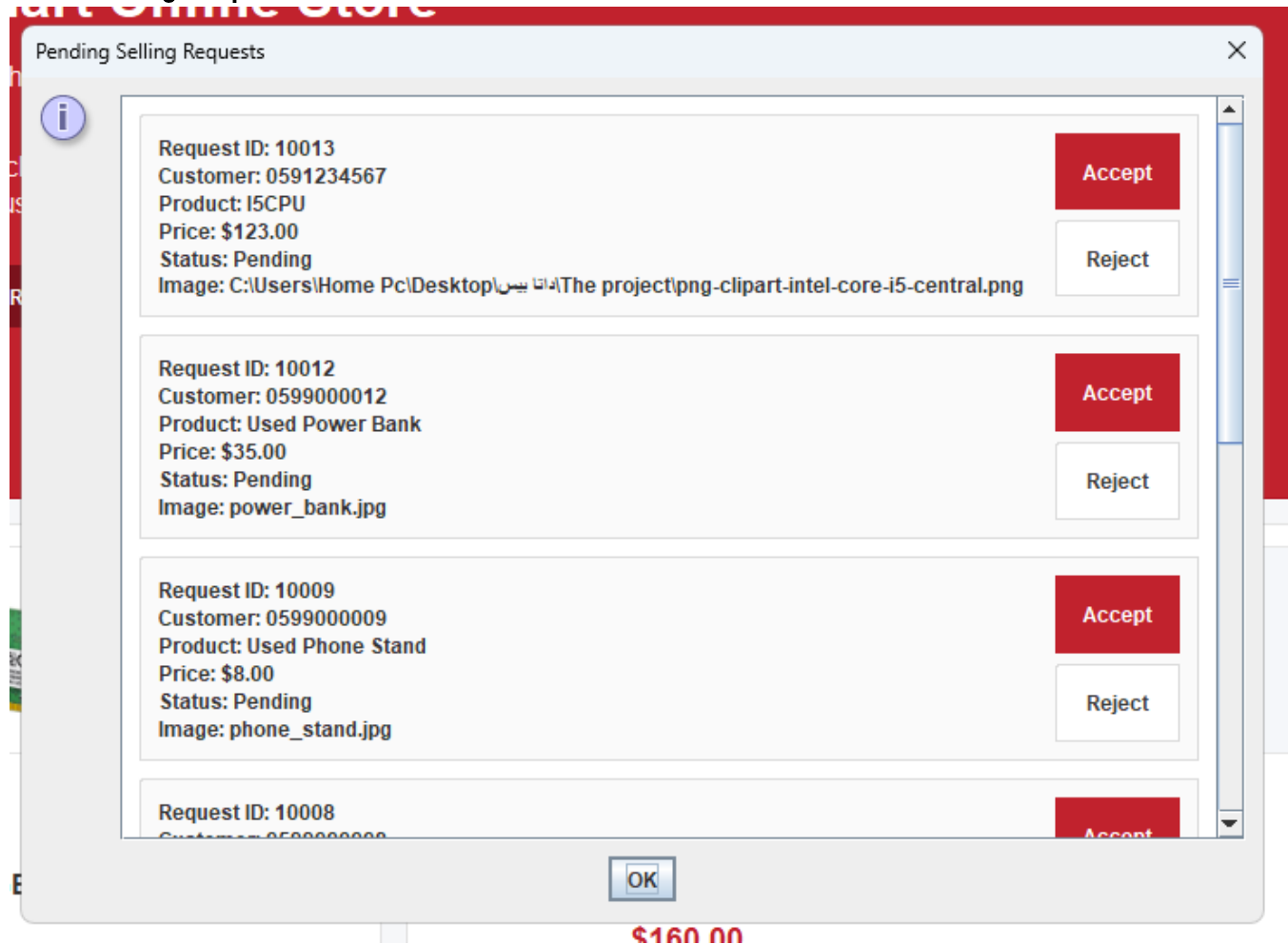


**View All products, it helps the user when he wants to edit something or delete:**



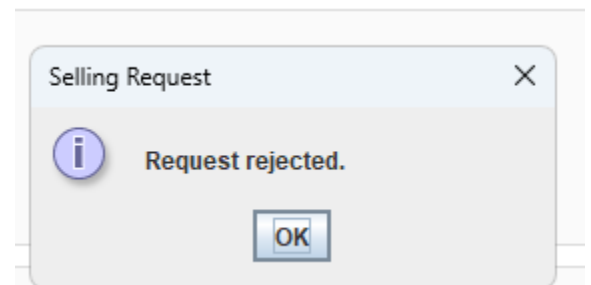
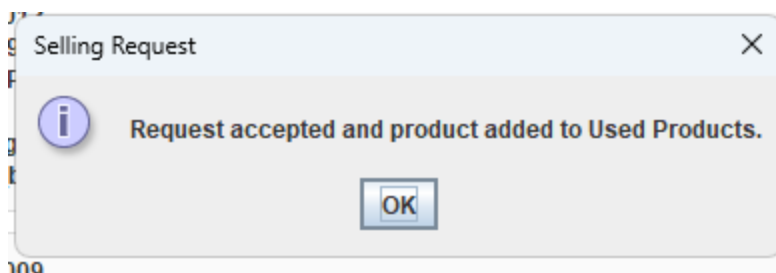
**View All Orders, helps the admin to check on all the orders:**

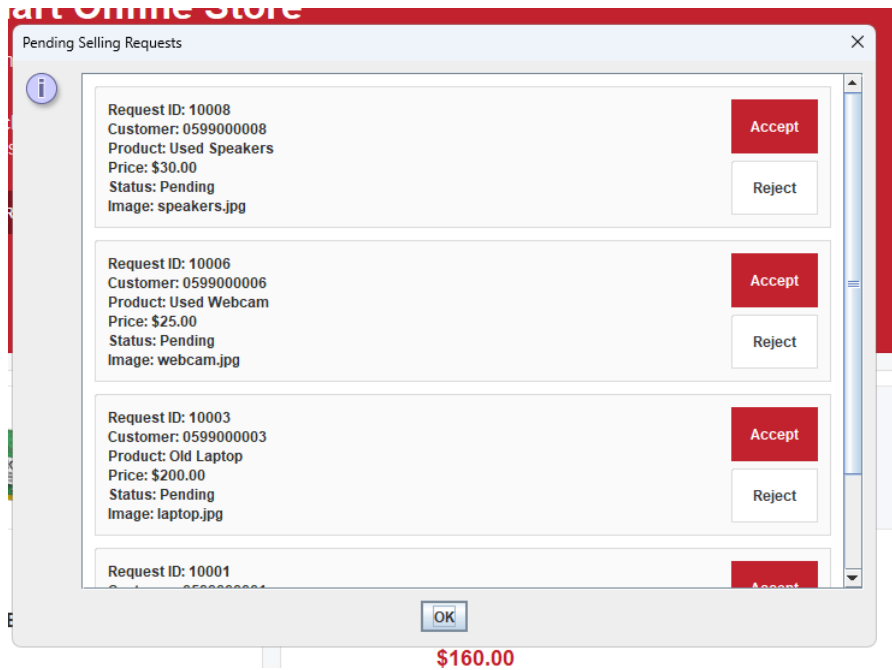


**View Selling Request:**

When we accept the Request it should appear on the Used category and Home, and the ones that are rejected should not appear on the Used category and Home.

So I am going to reject The Used Phone Stand, and accept i5CPU and Used Power Bank

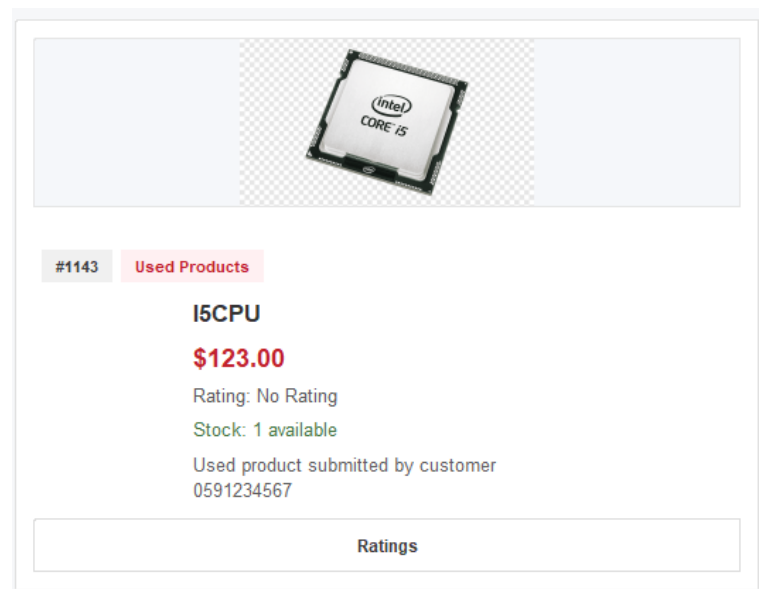
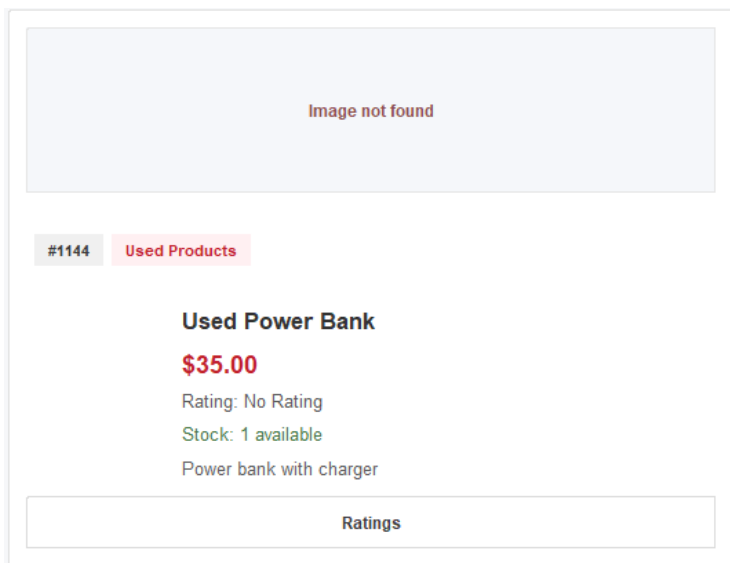




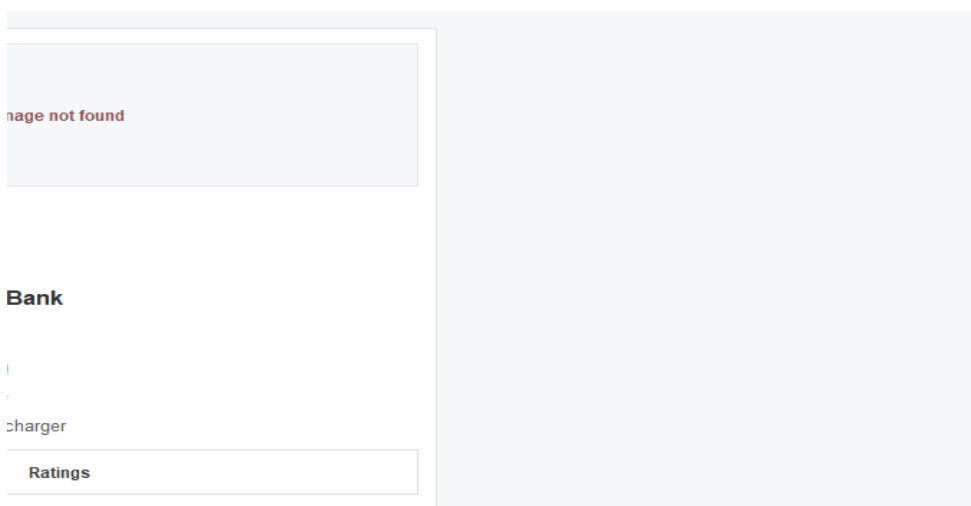
\$160.00

Accepted:

Used Power Bank does not have photo because of the wrong directory



Rejected: it is not next to Used Power Bank





We will now see the benefit of the View All Products  
Add Product:

Product Name:

Description:

Price:

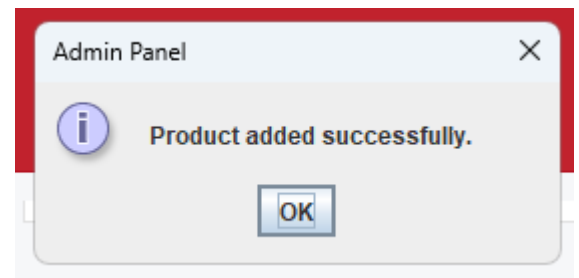
Quantity:

Category: Computer Parts

Discount: ☐ On Sale / Discount

Product Image:  No image selected

everything here is compulsory except  
image



After filling all the information we press OK and check if the product is added:

Product Name: Plastic Ball

Description: Plastic Soccer ball (9 inch) \_3 colors

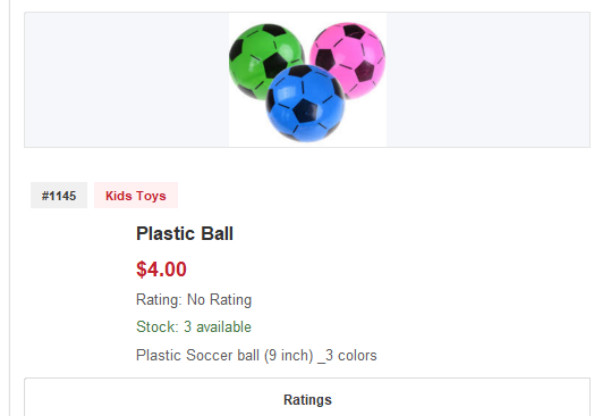
Price: 4

Quantity: 3

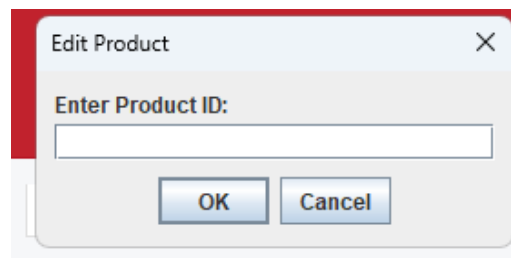
Category: Kids Toys

Discount: ☐ On Sale / Discount

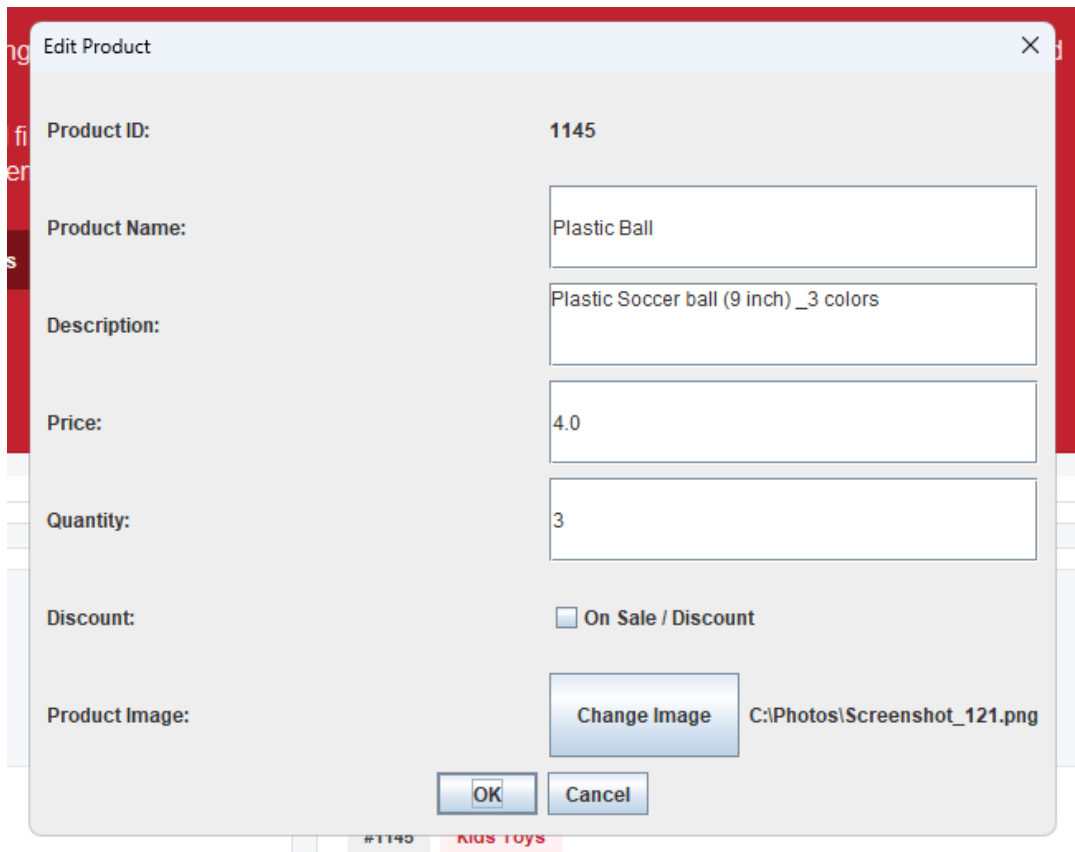
Product Image:  Screenshot\_121...



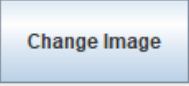
**Edit Product:**  
**We need to enter the id of the Product:**  
**Let's say the Plastic ball #1145**



Dialog box titled "Edit Product" with a close button (X). It contains a label "Enter Product ID:" followed by a text input field. Below the input field are "OK" and "Cancel" buttons.

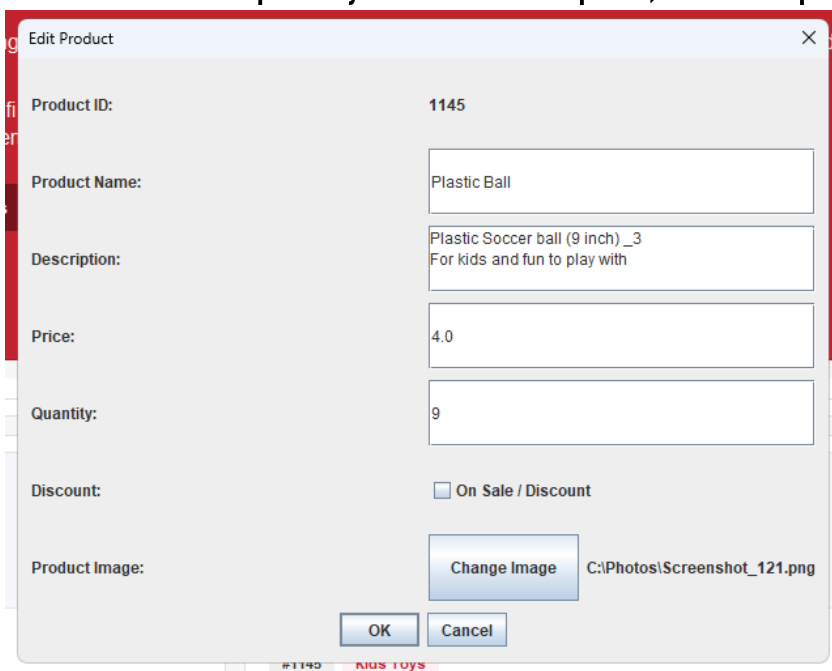


Dialog box titled "Edit Product" with a close button (X). It contains the following fields and values:

- Product ID: 1145
- Product Name: Plastic Ball
- Description: Plastic Soccer ball (9 inch) \_3 colors
- Price: 4.0
- Quantity: 3
- Discount: ☐ On Sale / Discount
- Product Image:  C:\Photos\Screenshot\_121.png

At the bottom are "OK" and "Cancel" buttons.

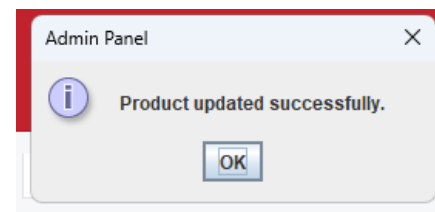
**Lets edit the quantity and the Description, and then press OK:**



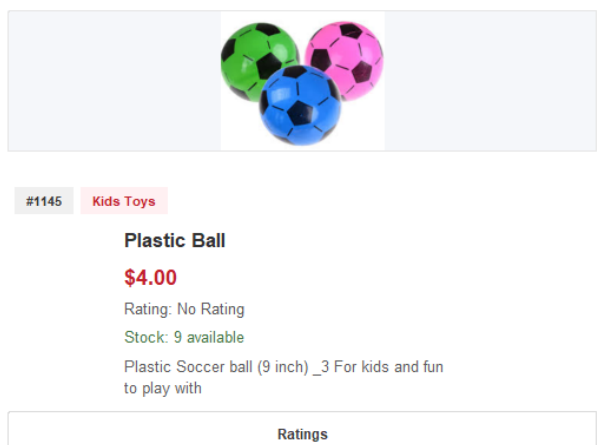
Dialog box titled "Edit Product" with a close button (X). It contains the following fields and values:

- Product ID: 1145
- Product Name: Plastic Ball
- Description: Plastic Soccer ball (9 inch) \_3 For kids and fun to play with
- Price: 4.0
- Quantity: 9
- Discount: ☐ On Sale / Discount
- Product Image:  C:\Photos\Screenshot\_121.png

At the bottom are "OK" and "Cancel" buttons.



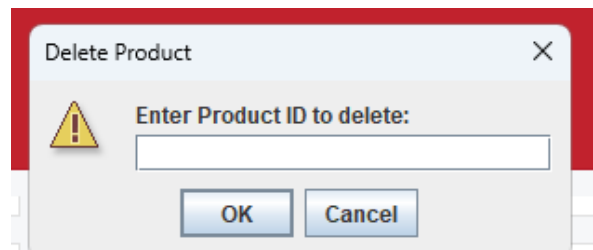
Dialog box titled "Admin Panel" with a close button (X). It contains an information icon (i) and the text "Product updated successfully." Below the text is an "OK" button.



Product listing for "Plastic Ball" (ID #1145, Category Kids Toys). It features an image of three soccer balls (green, blue, and pink) and the following details:

- Plastic Ball**
- \$4.00**
- Rating: No Rating
- Stock: 9 available
- Plastic Soccer ball (9 inch) \_3 For kids and fun to play with
- Ratings

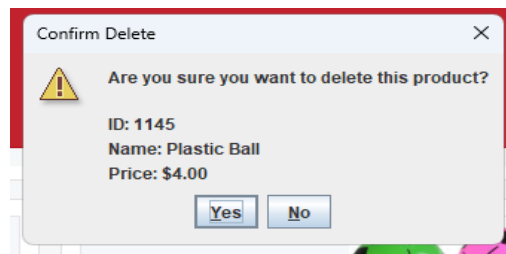
**Delete Product:**  
 We need to enter the id of the Product:  
 Let's say the Plastic ball #1145



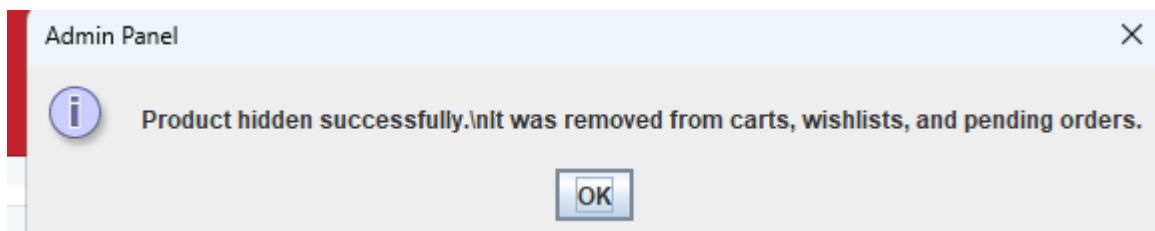
Actually, we don't really delete it from the Database we make it from the state Available to Hidden State

```
CREATE TABLE onlinestore.Product (
  product_id INT,
  product_name VARCHAR(150) NOT NULL,
  description VARCHAR(255),
  price NUMERIC(10,2) NOT NULL,
  image_url VARCHAR(255),
  discount BOOLEAN DEFAULT 0,
  availability_status VARCHAR(20) DEFAULT 'Available',
  category_id INT NOT NULL,
  stock_quantity INT NOT NULL DEFAULT 10,
  created_at TIMESTAMP DEFAULT now(),
  PRIMARY KEY (product_id),
  FOREIGN KEY (category_id) REFERENCES onlinestore.Category(category_id),
  CHECK (price >= 0),
  CHECK (stock_quantity >= 0),
  CHECK (availability_status IN 'Available', 'OutOfStock', 'Hidden'))
);
```

Then you need to confirm by clicking Yes



Then:



Before Deleting :



#1122 Kids Toys  
**Kids Doctor Play Set**  
**\$20.00**  
Rating: No Rating  
Stock: 10 available

Price: \$4.00



#1145 Kids Toys  
**Plastic Ball**  
**\$4.00**  
Rating: No Rating  
Stock: 9 available  
Plastic Soccer ball (9 inch) \_3 For kids and fun

After Deleting:



#1122 Kids Toys  
**Kids Doctor Play Set**  
**\$20.00**  
Rating: No Rating  
Stock: 10 available  
Children doctor toy set for pretend play

END OF THE GUI

## 15. Main User Workflow

The customer workflow starts with registration or login. After logging in, the customer can browse products, search and filter items, add products to the cart or wishlist, and complete checkout. During checkout, receiver information and payment method are entered, then the order is saved in the database with its related items.

After placing an order, the customer can view order details, confirm shipment, rate products or the website when allowed, submit selling requests, and use support chat when help is needed.

Main customer actions include:

- Register or log in.
- Browse, search, and filter products.
- Add products to cart or wishlist.
- Checkout and create an order.
- View and confirm orders.
- Rate products and the website.
- Submit selling requests.
- Use support chat.

## 16. Admin Workflow

The admin workflow focuses on managing and monitoring the store. After logging in with admin credentials, the admin can access management features that are not available to normal customers. Customer-only actions such as cart, wishlist, checkout, and selling requests are restricted for the admin.

Main admin actions include:

- View system statistics.
- View, add, edit, and hide products.
- Check product stock levels.
- View all customer orders.
- View product and website reviews.
- Accept or reject selling requests.
- Reply to customers through chat.
- Send a broadcast message to all customers.

## 17. Important Implemented Features

### **Stock Validation:**

The system checks stock before adding products to the cart and again before checkout. This prevents customers from ordering more than the available quantity.

### **Phone and City Validation:**

Phone numbers must follow the accepted format, and cities are selected from a fixed list. This keeps customer data more consistent.

### **Admin Restrictions:**

The admin cannot use customer-only features such as cart, wishlist, checkout, or selling requests.

### **Review Control:**

Customers can review products only after buying them and reaching the correct order status. This helps prevent fake reviews.

**Selling Requests:**

Customers can submit used products for sale. Pending requests can be edited or deleted, while the admin can approve or reject them.

**Chat Support:**

Customers can chat with the admin. The admin can reply to one customer or send a broadcast message to all customers.

**Image Support:**

Product images are loaded from stored image paths and displayed inside product cards.

## 18. Testing and Validation

Testing was performed manually by using the application as both customer and admin. The goal was to check that the GUI, database operations, validation rules, and relationships work correctly. The following table summarizes important test cases.

| Test Case                 | Action                                                 | Expected Result                             |
|---------------------------|--------------------------------------------------------|---------------------------------------------|
| Run Application           | Start Java application                                 | GUI opens and database connection succeeds. |
| Register Valid Customer   | Enter valid name, prefix, 7 digits, city, and password | Customer is created and logged in.          |
| Register Invalid Name     | Enter name with numbers                                | System shows name validation error.         |
| Login Invalid Phone       | Enter phone not starting with 059/056                  | System rejects the phone number.            |
| Add to Cart Without Login | Click Add to Cart while logged out                     | System asks user to log in first.           |
| Add to Cart Beyond Stock  | Add quantity greater than available stock              | System shows Stock Error.                   |
| Checkout Valid Cart       | Enter valid receiver details                           | Order is created and cart is cleared.       |
| Checkout Changed Stock    | Product becomes out of stock before checkout           | Item is removed or quantity adjusted.       |
| Admin Add Product         | Enter valid product data                               | Product appears in product list.            |
| Admin Edit Product        | Update price/quantity/image                            | Product details are updated.                |
| Admin Delete Product      | Hide an existing product                               | Product disappears from customer view.      |
| Chat Customer to Admin    | Customer sends message                                 | Admin can see and reply.                    |
| Admin Broadcast           | Admin selects All and sends message                    | All customers receive message.              |
| Selling Request           | Customer submits used product                          | Admin can accept or reject request.         |

## Appendix Sample Testing Checklist

Database connection works

Product images appear

Search works

Category navigation works

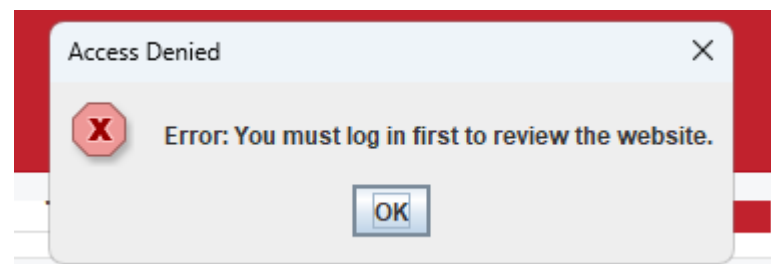
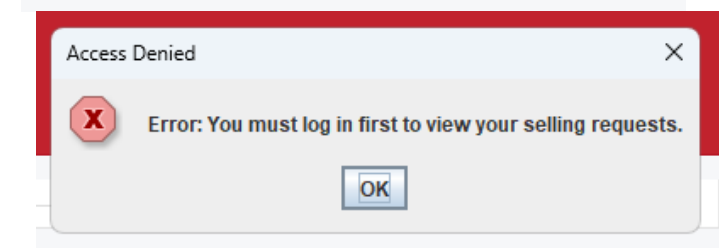
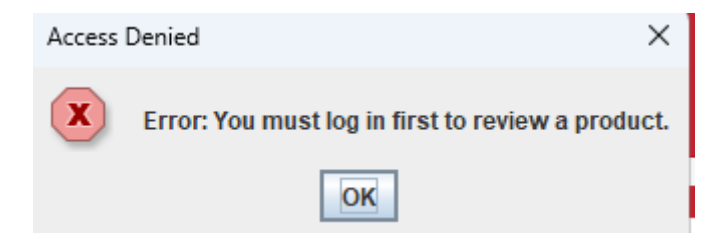
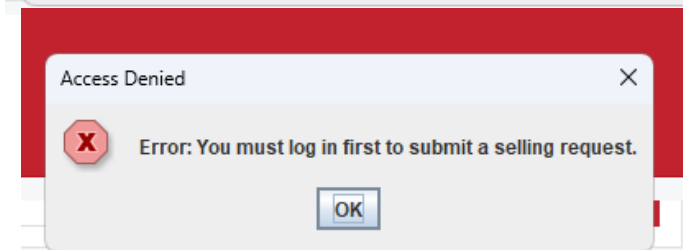
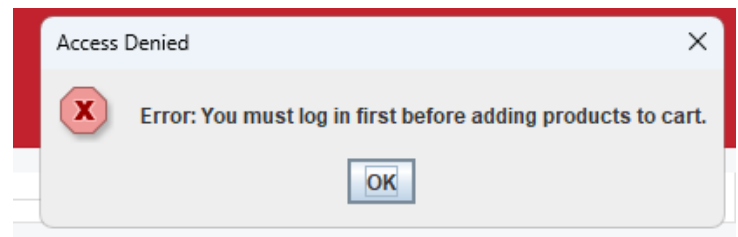
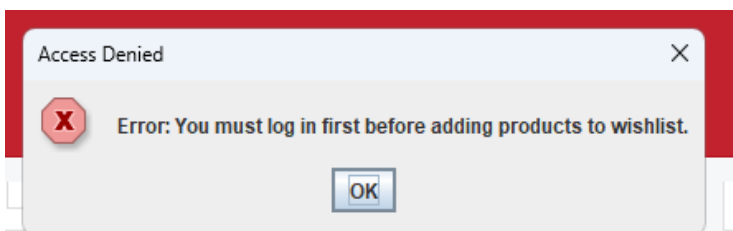
Sort by rating works

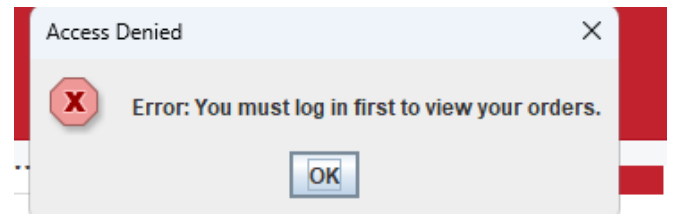
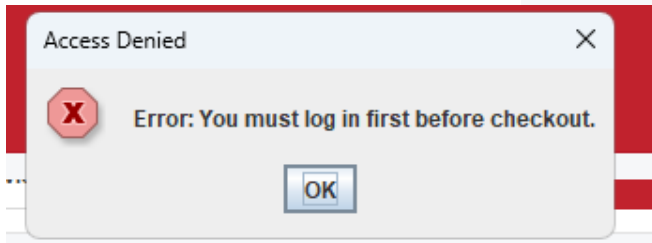
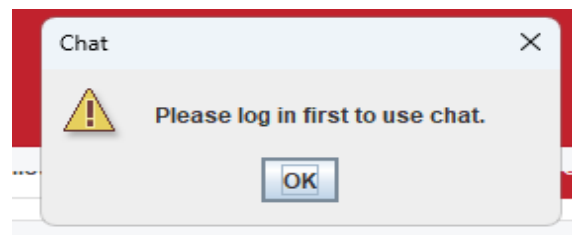
Sort by stock works

Sort by price works

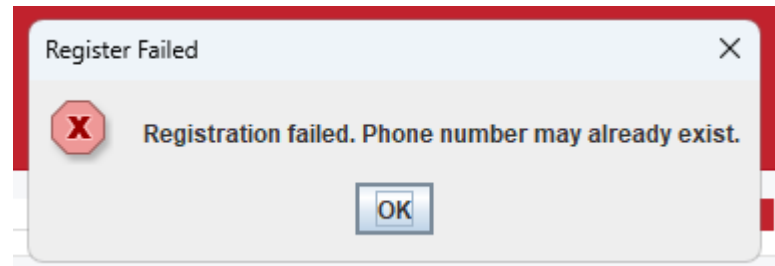
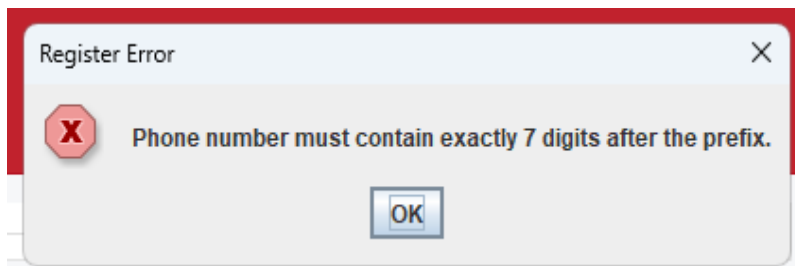
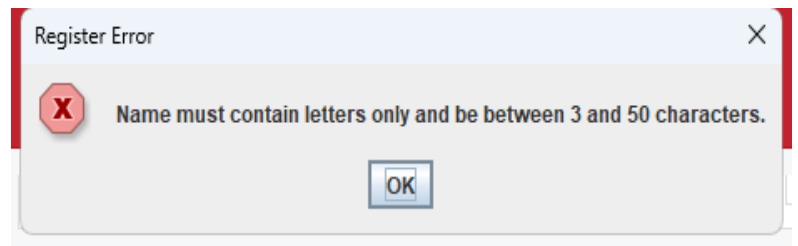
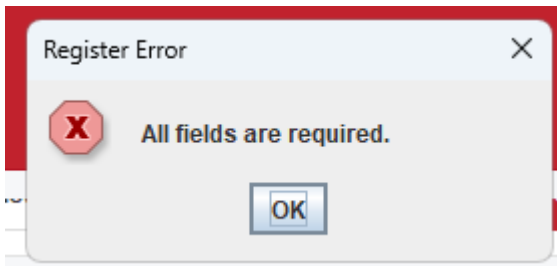
Register valid customer works  
Register invalid phone rejected  
Register invalid name rejected  
Login customer works  
Login admin works  
Logout clears cart view  
Add to cart works  
Add beyond stock rejected  
Wishlist works  
Checkout creates order  
Order confirmation works  
Card validation works  
Product review rules work  
Website review works  
Selling request works  
Admin accepts request  
Admin rejects request  
Chat user/admin works  
Admin broadcast works  
Clean and Build succeeds

### GUI Messages Before Register/Login:

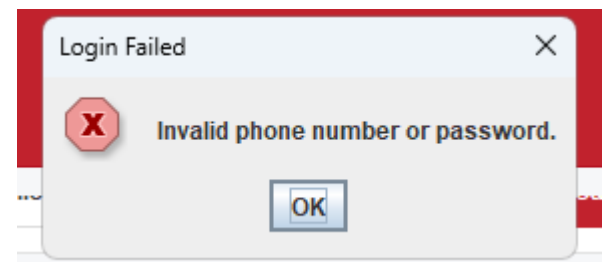
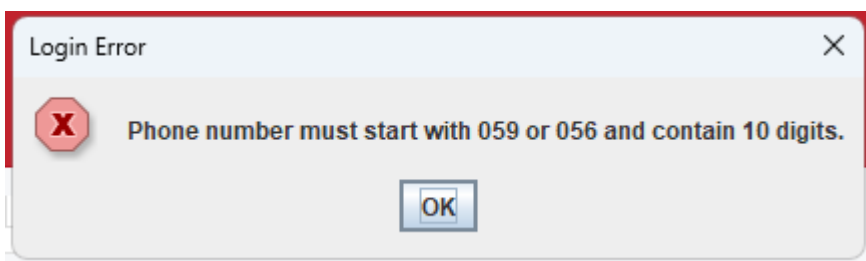




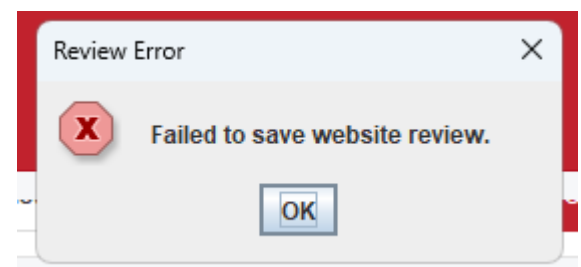
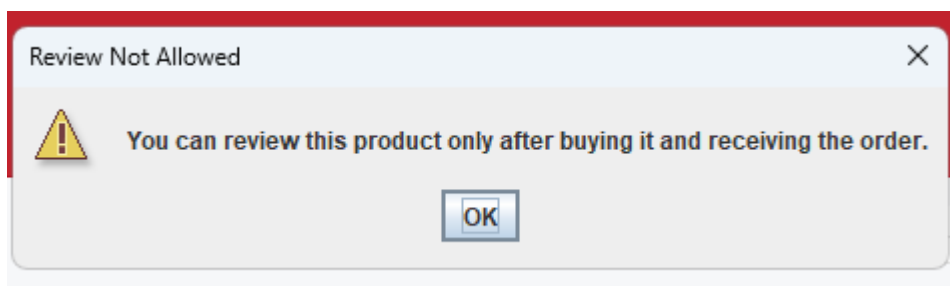
### Wrong Register Inputs:



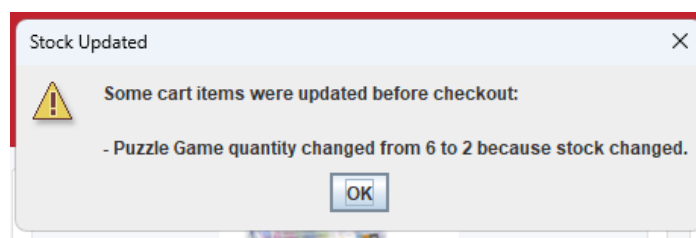
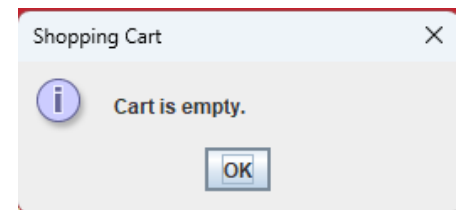
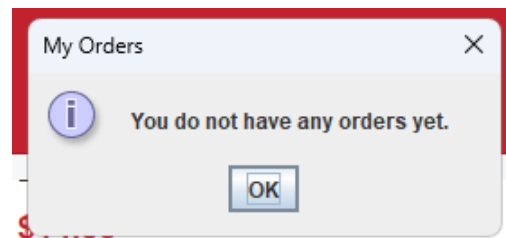
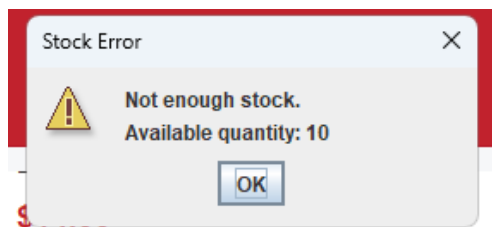
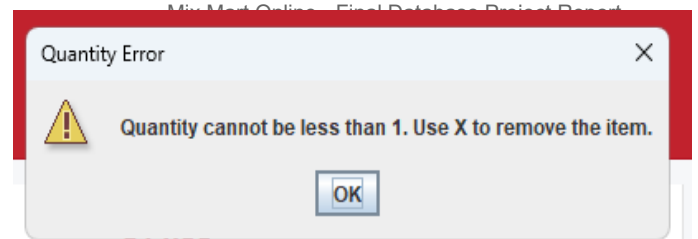
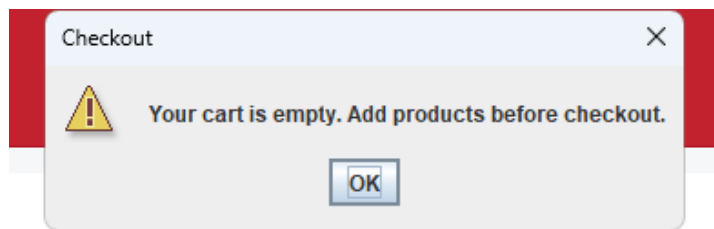
### Wrong login Inputs:



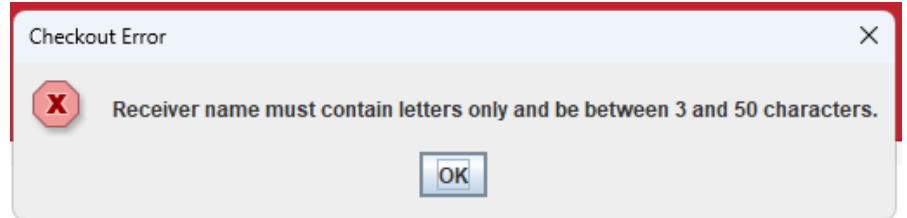
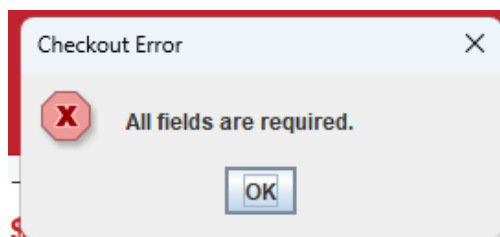
### After Logging Warning/Error Messages:



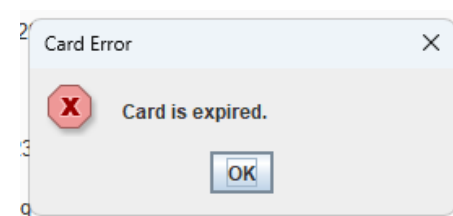
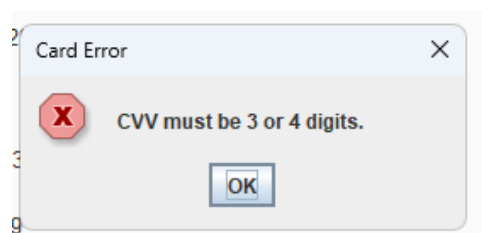
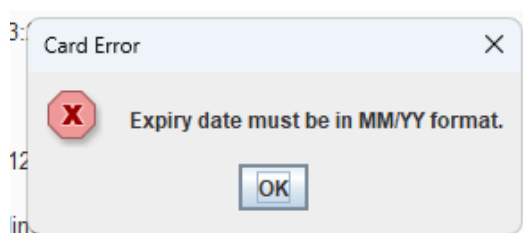
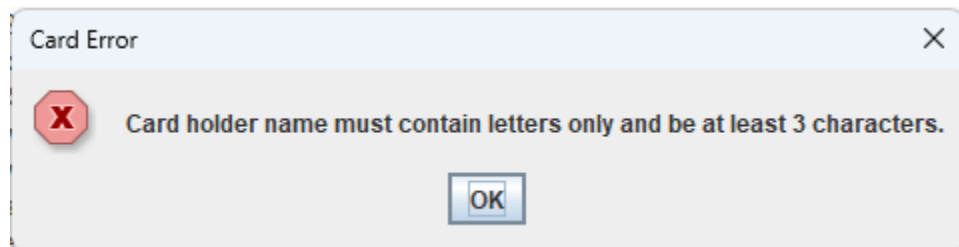




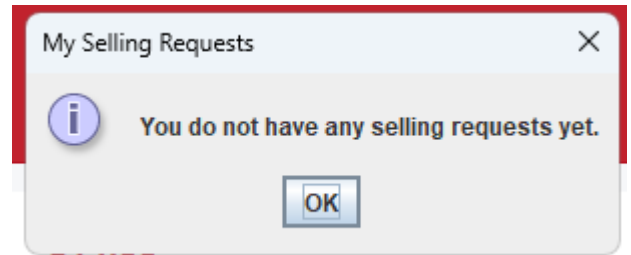
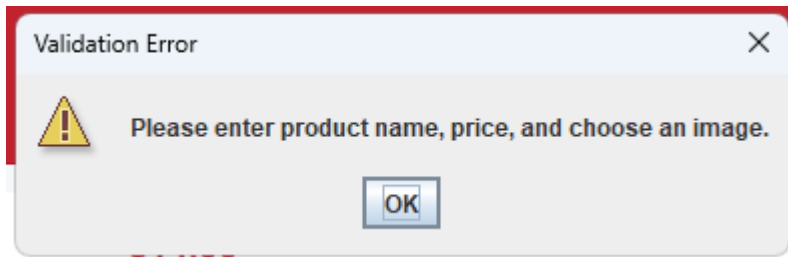
### Checkout Information Error Messages:



### Order Card Information Error Messages:

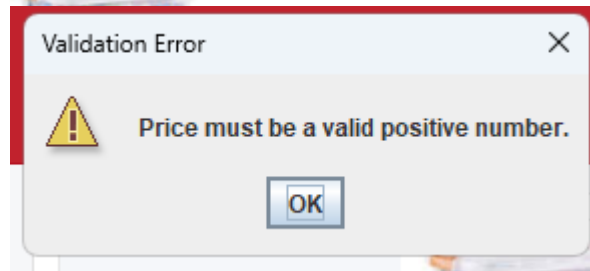
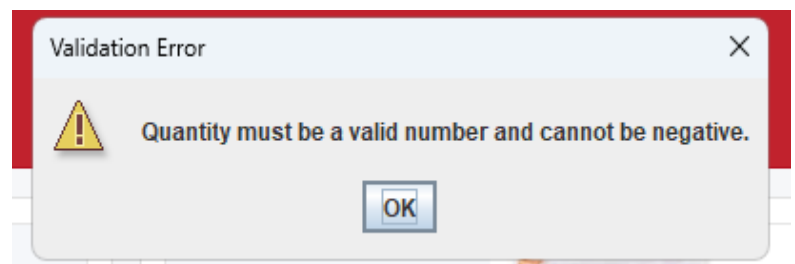
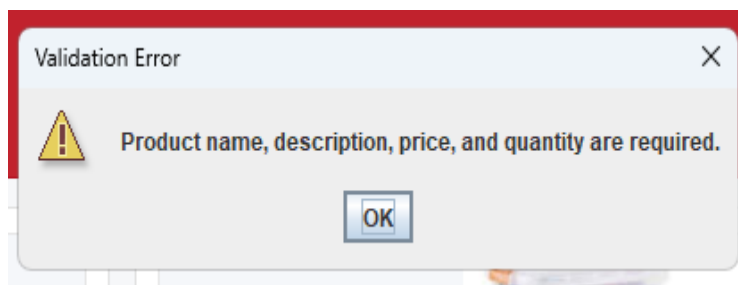


### Show your item/My Selling Error Messages:



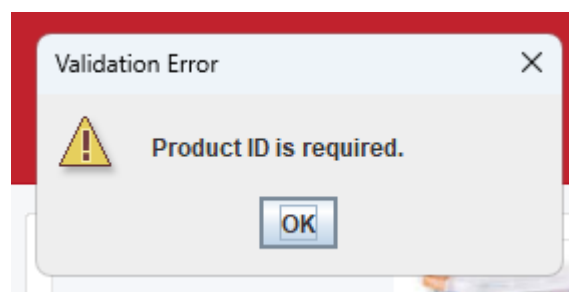
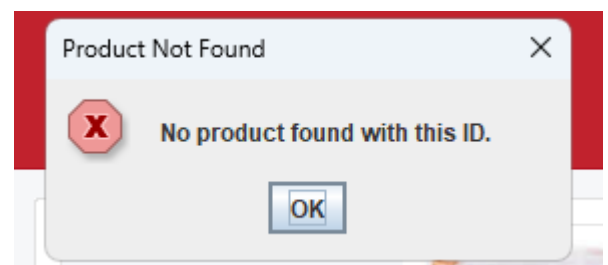
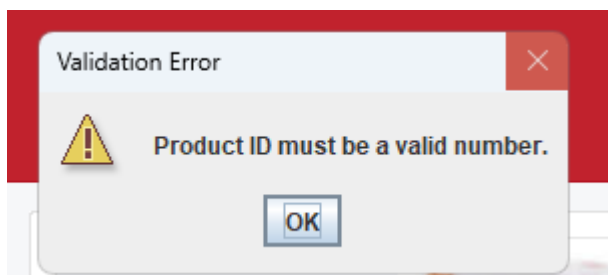
### Admi Panel:

#### Add products Error Messages:



#### Edit/Delete Product Error Messages:

Have the same error Messages as the add Product



## 19. Error Handling and Data Validation

The application includes validation rules to reduce input errors and protect the database from invalid data. Some rules are handled in Java, while other rules are supported by SQL constraints.

The system validates:

- Names: letters and spaces only.
- Phone numbers: accepted prefix and correct length.
- Prices: positive values only.
- Quantities: stock cannot be negative.
- Cart quantity: cannot exceed available stock.
- Card information: valid card number, expiry date, and CVV.
- Status values: controlled using SQL CHECK constraints where needed.

These rules improve data quality and help avoid problems during registration, checkout, product management, and order processing.

## 20. Assets and Submitted Files

The final submission includes the Java source code, SQL script, database dump if required, ER diagram, normalization document, product images folder, PowerPoint presentation, project report, and demo video. These files support the final delivery and help demonstrate both the database design and the working application.

## 21. Limitations and Future Work

Although the project includes many features, it is still a student-level system and can be improved in the future.

Possible improvements include:

- Using password hashing instead of plain text passwords.
- Moving from Java Swing to a web-based system.
- Storing images on a server or cloud storage instead of local paths.
- Adding a separate Admin table with stronger role-based permissions.
- Connecting payment to a real payment gateway.
- Adding automatic chat refresh.
- Adding more advanced dashboard charts and reports.

## 22. Relationship and Cardinality Explanation

The following table explains the main relationships in the database. Cardinality is important because it controls how many rows in one table can be connected to rows in another table.

| Relationship            | Cardinality | Explanation                                                                |
|-------------------------|-------------|----------------------------------------------------------------------------|
| Customer - ShoppingCart | 1 : 1       | Each customer has one main cart. ShoppingCart has a UNIQUE customer_phone. |
| Customer - Orders       | 1 : N       | A customer can place many orders, but each order belongs to one customer.  |
| Orders - OrderItem      | 1 : N       | One order contains many order items.                                       |

|                                 |       |                                                    |
|---------------------------------|-------|----------------------------------------------------|
| Product - OrderItem             | 1 : N | One product can appear in many order items.        |
| ShoppingCart - ShoppingCartItem | 1 : N | One cart can contain many cart items.              |
| Product - ShoppingCartItem      | 1 : N | One product can appear in many carts.              |
| Category - Product              | 1 : N | One category has many products.                    |
| Orders - Payment                | 1 : 1 | Each order has one payment record.                 |
| Orders - Shipment               | 1 : 1 | Each order has one shipment record.                |
| Customer - Review               | 1 : N | A customer can write many reviews.                 |
| Product - Review                | 1 : N | A product can receive many product reviews.        |
| Customer - Wishlist             | 1 : N | A customer can save many wishlist items.           |
| Product - Wishlist              | 1 : N | A product can be saved by many customers.          |
| Customer - SellingRequest       | 1 : N | A customer can submit many selling requests.       |
| Customer - ChatMessage          | 1 : N | A customer can have many chat messages with admin. |

## 23. Detailed Data Dictionary

The data dictionary describes the main columns in each table and the reason for using them. This section helps connect the ERD with the SQL implementation.

### Customer

| Column            | Type         | Description                       |
|-------------------|--------------|-----------------------------------|
| phone_number      | VARCHAR(20)  | Primary key and login identifier. |
| City              | VARCHAR(15)  | Customer city.                    |
| Name              | VARCHAR(20)  | Customer full name.               |
| customer_password | VARCHAR(100) | Password used for login.          |
| created_at        | TIMESTAMP    | Account creation time.            |

### Product

| Column              | Type          | Description                       |
|---------------------|---------------|-----------------------------------|
| product_id          | INT           | Primary key.                      |
| product_name        | VARCHAR(150)  | Product display name.             |
| description         | VARCHAR(255)  | Product details shown in GUI.     |
| price               | NUMERIC(10,2) | Product price.                    |
| image_url           | VARCHAR(255)  | Local image path.                 |
| discount            | BOOLEAN       | Indicates if product is on sale.  |
| availability_status | VARCHAR(20)   | Available, OutOfStock, or Hidden. |
| category_id         | INT           | Foreign key to Category.          |
| stock_quantity      | INT           | Available quantity.               |

### Orders

| Column         | Type        | Description              |
|----------------|-------------|--------------------------|
| order_id       | INT         | Primary key.             |
| customer_phone | VARCHAR(20) | Foreign key to Customer. |

|                |               |                                 |
|----------------|---------------|---------------------------------|
| total_amount   | NUMERIC(10,2) | Total order price.              |
| order_date     | TIMESTAMP     | Date and time of order.         |
| order_status   | VARCHAR(20)   | Pending, Shipped, or Delivered. |
| receiver_name  | VARCHAR(100)  | Delivery receiver.              |
| receiver_phone | VARCHAR(20)   | Receiver phone number.          |
| city           | VARCHAR(100)  | Delivery city.                  |
| street         | VARCHAR(255)  | Delivery street/address.        |

## Review

| Column         | Type         | Description                                |
|----------------|--------------|--------------------------------------------|
| review_id      | INT          | Primary key.                               |
| customer_phone | VARCHAR(20)  | Reviewer customer.                         |
| product_id     | INT NULL     | Product being reviewed, if product review. |
| order_id       | INT NULL     | Related order.                             |
| rating         | INT          | Rating between 1 and 5.                    |
| comment        | VARCHAR(255) | Review text.                               |
| review_date    | TIMESTAMP    | Review date.                               |
| review_type    | VARCHAR(20)  | Product or Website.                        |

## ChatMessage

| Column         | Type               | Description            |
|----------------|--------------------|------------------------|
| message_id     | INT AUTO_INCREMENT | Primary key.           |
| customer_phone | VARCHAR(20)        | Conversation customer. |
| sender_type    | VARCHAR(20)        | Customer or Admin.     |
| message_text   | VARCHAR(1000)      | Message body.          |
| sent_at        | TIMESTAMP          | Message time.          |

## 24. SQL Constraints and Integrity Rules

The project database uses constraints to protect data integrity. These rules are important because they do not depend only on the Java application. Even if data is inserted directly through SQL, the database can still reject invalid or unrelated records.

Primary keys are used to uniquely identify records in each table, such as `product_id`, `order_id`, `payment_id`, and `review_id`. Foreign keys connect related tables and help prevent orphan records, such as linking `Product.category_id` with `Category.category_id` and `OrderItem.order_id` with `Orders.order_id`.

The database also uses unique constraints to prevent duplicated relationships, such as adding the same product more than once to the same wishlist. Check constraints are used to control accepted values, such as rating values, product availability status, order status, payment status, and request status. Default values are used for dates and initial statuses, such as `created_at` using the current timestamp and `cart_status` starting as `Active`.

These constraints help keep the database consistent and support the validation rules used in the Java application.

## 25. Conclusion

Mix Mart Online is a practical online shopping system that connects database concepts with a working Java Swing application. Through this project, we applied important topics from the Database Systems course, including ER modeling, relational schema design, primary keys, foreign keys, normalization, SQL DDL and DML statements, constraints, and JDBC connectivity.

The system includes both customer and admin features. Customers can register, log in, browse products, search and filter items, use the cart and wishlist, place orders, choose payment and delivery options, write reviews, submit used products for sale, and chat with the admin. The admin can manage products, view orders, check statistics, read reviews, handle selling requests, and reply to customer messages.

Overall, the project gave us practical experience in designing, implementing, testing, and presenting a database-backed system. It helped us understand how good database design reduces repeated data, keeps information organized, and supports real application operations.

**REPORT IS DONE.**